Seeing, Hearing, and Acting: A Survey of Audio-Visual Collaboration for Embodied Intelligence

Zhonghong Ou, Yifan Zhu, Pan Hui, *Fellow, IEEE,*

***Abstract*—Audio-Visual Embodied Artificial Intelligence (AV-EAI) has emerged as an indispensable frontier driven by the recent convergence of World Models, Physical AI, and omni-modal foundation models, utilizing vision and audio as the essential physical perception units to ground intelligent agents in the real world. While previous studies have explored isolated, task-specific audio-visual applications or general multi-modal learning, the highly diverse audio-visual literature has not been systematically reviewed in a unified manner, leaving taxonomies and evaluation protocols in a chaotic state. To bridge this gap, this survey establishes a unified taxonomy to systematically characterize how audio-visual capabilities evolve from sensory alignment and scene perception toward agent-centric reasoning and embodied interaction. Specifically, we organize the AV-EAI landscape into three progressive capability dimensions: Perception, Reasoning, and Interaction, tracing the evolution from multi-sensory correspondence and scene understanding to situated reasoning and closed-loop embodied interaction. Under this framework, we first examine how complementary audio-visual signals support cross-modal consistency, grounding, event analysis, and physical scene understanding (Perception). We then investigate how specialized models, omni-modal foundation models, and emerging agentic systems progressively enable cross-modal, spatial, physical-causal, multi-step, and context-aware reasoning (Reasoning). Finally, we survey how these perceptual and reasoning capabilities are translated into closed-loop embodied behaviors across mobility, manipulation, communication, and increasingly generalist audio-visual-language policies (Interaction). Furthermore, we systematically consolidate the scattered datasets, simulators, and evaluation metrics, establishing a standardized foundation for the AV-EAI community. Finally, we critically examine pressing open challenges—including sim-to-real transfer, real-time sensory fusion latency, and the development of large Vision-Language-Action-Audio (VLAA) models. Ultimately, by demystifying these critical directions, we aspire for this survey to serve as a vital signpost and a strategic roadmap for the research community, guiding the next leap in Audio-Visual Embodied AI. To facilitate ongoing research, a comprehensive and continuously updated repository summarizing the surveyed literature and benchmarks is publicly available at https://github.com/Raiden-cn/Awesome_AVEI.**

***Index Terms*—Audio-Visual, Embodied Intelligence, Physical AI, World Model, Omni-modal**

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I. Introduction

"Sight reveals the differences of things; but hearing brings us the meanings"

— **Immanuel Kant**, *Critique of Pure Reason*

The evolution of Artificial General Intelligence (AGI) and Physical AI has inexorably shifted the frontier of machine learning from passive digital computation to active engagement within the physical world. To navigate and manipulate complex environments autonomously, intelligent agents must rely on Audio-Visual as their primary and highly complementary sensory channels. While vision provides high-resolution spatial geometry, it is inherently bounded by the line-of-sight and highly vulnerable to occlusions and illumination variations. Audition, conversely, creates an omnidirectional perceptual sphere and captures rich, physics-grounded feedback triggered by physical interactions. Consider a domestic robot operating in a bustling household: it cannot visually perceive a boiling kettle around a blind corner, nor can it accurately ascertain the hollowness of an opaque container merely by looking at it; however, the acoustic cues of whistling steam or the impact sound of a tap instantly reveal these hidden states. Today, the unprecedented breakthroughs in omni-modal foundation models—exemplified by natively unified architectures (e.g., GPT-4o [1], Gemini [2]) and emerging Vision-Language-Action-Audio (VLAA) paradigms—have finally endowed agents with the cognitive capacity to process these synchronized multi-sensory streams in real time. This profound convergence has catalyzed a critical paradigm shift, driving the research community toward the era of Audio-Visual Embodied Artificial Intelligence (AV-EAI), in which agents must not only perceive multimodal environments, but also reason about them and interact with them in closed loop.

Despite the profound potential of utilizing audio-visual modalities, deploying AV systems in physical embodied agents confronts formidable challenges. First, the inherent heterogeneity between spatial-heavy visual signals and temporal-heavy audio streams complicates the design of unified representations, particularly under real-time constraints where latency is critical for closed-loop motor control. Second, embodied agents operate in highly dynamic and noisy physical environments; distinguishing target acoustic cues (e.g.,

human commands or shattering glass) from background cacophony, while maintaining robust tracking under severe visual occlusions, remains notoriously difficult. Finally, a critical sim-to-real gap persists. While visual rendering in simulators (e.g., Habitat, AI2-THOR) has reached photorealism, high-fidelity acoustic rendering—accurately modeling sound diffraction, reverberation, and material-dependent acoustic physics—remains computationally prohibitive, severely hindering the transferability of learned AV policies to the real world.

To tackle these bottlenecks, recent years have witnessed a paradigm shift driven by advanced generative architectures and foundation models. Transformer-based models [**?**], [**?**] have revolutionized cross-modal alignment, employing spatio-temporal attention mechanisms to construct robust, shared latent spaces that fuse vision and audio more effectively than traditional late-fusion methods. Concurrently, the emergence of Diffusion models [**?**], [**?**] has provided powerful tools for tackling the sim-to-real data scarcity. By synthesizing high-fidelity, physics-grounded audio-visual environments and generating realistic multi-modal sensory data, diffusion techniques have significantly augmented training pipelines. Furthermore, the rapid convergence of large language models (LLMs) with multi-modal inputs—resulting in audio-native or omni-modal foundation models like Qwen-Audio [3], ImageBind [4], and GPT-4o [1]—has endowed agents with unprecedented zero-shot reasoning capabilities across diverse AV inputs.

Among these rapid algorithmic developments, several trajectories show exceptional promise for the future of Embodied AI. The transition towards Vision-Language-Action-Audio (VLAA) architectures is arguably the most critical trend, as it closes the loop from passive perception directly to physical action, utilizing language as a semantic bridge. Additionally, Active Acoustic Sensing [**?**]—where agents purposefully emit sounds (e.g., tapping, echolocation) to probe material properties or spatial geometries that are visually ambiguous—is emerging as a highly promising avenue for robust robotic manipulation. Furthermore, Physics-Informed AV Learning [**?**], which explicitly embeds acoustic wave equations and physical constraints into neural architectures, is showing great potential in closing the sim-to-real gap.

Existing surveys provide complementary perspectives on audio-visual intelligence, with their coverage of perception, reasoning, interaction, foundation models, and embodiment compared in Table **??**. General reviews organize the field through task families, cognitive principles, or joint audio–visual analysis [5], [6], [7], while focused reviews examine self-supervised representations, speaker tracking, navigation, segmentation, and music question answering [8], [9], [10], [11], [12]. Recent foundation-model surveys also encompass reasoning and embodied interaction alongside perception and generation [13]. These perspectives motivate a complementary, embodiment-centered synthesis organized around the functional connections among perception, reasoning, and interaction: how sensory evidence supports inference, how inference guides embodied behavior, and how interaction generates new evidence to refine perception and subsequent decisions.

Building on these perspectives, this survey examines how audio–visual capabilities jointly support an agent's understanding and interaction with its environment. Our main contributions are threefold: **1)** We establish a unified **Perception–Reasoning–Interaction** taxonomy based on principal outputs and functional roles, distinguishing enabling perceptual and reasoning capabilities from executed interaction. This framework connects how perception supplies evidence, reasoning guides decisions, and interaction generates feedback for subsequent understanding. **2)** We synthesize methodological developments across specialized methods, foundation models, and agentic systems, analyzing when and how auditory evidence complements vision in inference, active sensing, and action feedback across mobility, manipulation, and communication. **3)** We consolidate simulators, datasets, and evaluation benchmarks according to the capabilities they support, and identify shared limitations and research directions in evidence grounding, streaming inference, acoustic transfer, and closed-loop evaluation. Throughout, we emphasize when sound changes an agent's decisions and whether those changes improve embodied performance.

TABLE I
THE SCOPE OF RELATED SURVEYS ACROSS FIVE DIMENSIONS. ✓: DEDICATED DISCUSSION OF TASKS OR METHODS; ○: LIMITED DISCUSSION; -: NOT A FOCUS OF THE REVIEWED SCOPE.

Survey	Venue	Perception	Reasoning	Interaction	Foundation Model	Embodied
[5]	IJAC 2021	✓	-	-	-	-
[6]	arXiv 2022	✓	✓	✓	-	✓
[9]	arXiv 2023	✓	-	○	-	○
[7]	Access 2024	✓	✓	-	✓	-
[10]	I.F. 2024	✓	✓	✓	-	✓
[8]	Neur. 2025	✓	-	○	-	○
[11]	arXiv 2025	✓	○	-	✓	-
[12]	ACL 2026	✓	✓	-	✓	-
[13]	arXiv 2026	✓	✓	✓	✓	✓
Ours	-	✓	✓	✓	✓	✓

The remainder of this paper is organized as follows. Section II introduces the fundamental background and details the proposed PRIA taxonomy. Following this framework, Sections III, IV, and V systematically survey and discuss current methodologies and their corresponding benchmarks from three thematic angles: embodied perception and reasoning, active multi-sensory interaction, and contact-rich action execution. Section VI critically discusses open challenges and outlines promising future directions. Section VII concludes the survey.

II. BACKGROUND

A. Taxonomy

To characterize how audio–visual collaboration supports embodied intelligence, we adopt a capability-oriented hierarchy, as shown in Fig. 2. Its three top-level domains are distinguished by their principal outputs and functional roles. **Perception** estimates signals, entities, events, and spatial or acoustic structures from observations. **Reasoning** integrates evidence to derive relations, explanations, predictions, or plans. **Interaction** produces actions, sensing strategies, or communicative responses that participate directly in a physical, simulated, or communicative loop. Perception and Reasoning include both capabilities deployed in embodied agents and



Fig. 1. Overview of the three core capabilities of Audio-Visual Embodied Intelligence, illustrated by an autonomous-driving scenario. **Perception** identifies visual and auditory cues, including a green traffic light and an emergency siren. **Reasoning** integrates these cues to determine the need to yield. **Interaction** translates this decision into actions, while environmental feedback supports subsequent perception and reasoning.

enabling tasks with a clear connection to embodied understanding or decision-making.

The lower levels organize each domain according to its characteristic objectives and modes of operation. **Perception** comprises *Consistency*, *Grounding*, *Analysis*, and *Scene Understanding*, covering cross-modal compatibility, spatial and temporal evidence localization, recognition and source separation, and geometric and acoustic estimation, respectively. **Reasoning** comprises *Specialized-Model Reasoning*, addressing event, spatial, and physical–causal inference; *Foundation-Model Reasoning*, covering native omnimodal, multi-step, and long-context reasoning; and *Agentic Reasoning*, incorporating tools, memory, and multi-agent collaboration. **Interaction** comprises *Mobility Interaction*, *Manipulative Interaction*, and *Communicative Interaction*. *Generalist Policies* form a cross-cutting category concerned with policy reuse and generalization across interaction tasks.

This hierarchy combines functional distinctions at the top level with domain-specific organization below. In particular, the reasoning categories describe complementary paradigms rather than mutually exclusive architectures: foundation models can support both specialized tasks and agentic systems. Works spanning multiple capabilities are discussed according to their principal output and evaluation objective, with supporting functions identified where relevant. Architectural details, training strategies, datasets, and deployment settings provide additional descriptors. The connections among the three domains are functional rather than a strict chronological sequence: perception supplies evidence for reasoning, reasoning informs interaction, and interaction generates observations that can revise subsequent perception and inference.

B. Formal Definitions

To delimit the scope of this survey, we formalize audio–visual collaboration for embodied intelligence within a partially observed controlled dynamical system. At time step t, let $s_t \in \mathcal{S}$ denote the latent state of the environment and the agent, $o_t^V \in \mathcal{O}^V$ the visual observation, $o_t^A \in \mathcal{O}^A$ the auditory observation, and $u_t \in \mathcal{U}$ the action. We define the joint audio–visual observation as

$$o_t^{AV} := \left(o_t^V, o_t^A\right). \tag{1}$$

Audio–visual collaboration is formulated as a partially observed controlled dynamical system:

$$s_{t+1} \sim p_{\text{env}}\left(s_{t+1} \mid s_t, u_t\right), \qquad o_t^{AV} \sim p_{\text{env}}\left(o_t^V, o_t^A \mid s_t\right). \tag{2}$$

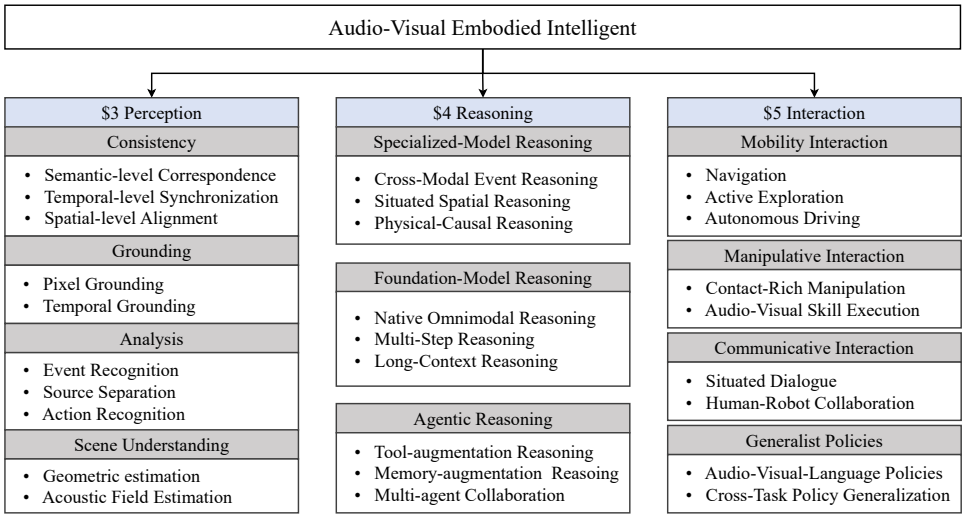


Fig. 2. Hierarchically-structured taxonomy of Audio-Visual Embodied Intelligence.

TABLE II
UNIFIED NOTATION USED IN OUR FORMALIZATION.

Symbol	Meaning
$s_t \in \mathcal{S}$	Latent state of the environment and agent at time t
$o_t^V \in \mathcal{O}^V$	Partial or noisy visual observation of s_t
$o_t^A \in \mathcal{O}^A$	Partial or noisy auditory observation of s_t
o_t^{AV}	Joint audio–visual observation (o_t^V, o_t^A)
$u_t \in \mathcal{U}$	Action or executable response of the agent at time t
$\ell \in \mathcal{L}$	Task instruction or goal
$q \in \mathcal{Q}$	Task-level perceptual or reasoning query
h_t	Audio–visual observation and interaction history
$x_t = g(s_t)$	Task-relevant perceptual state, signal, or structure
y_t^R	Task-level reasoning output
r_t	Optional reasoning trace

We denote the observable history by

$$h_t := \left(o_{1:t}^{AV}, u_{1:t-1}\right), \tag{3}$$

and use it as the available context under partial observability. For offline perception and reasoning tasks, the action history may be empty. A system is regarded as audio–visually collaborative when the relationship between auditory and visual information is functionally used to produce its primary output, rather than the two modalities being processed as unrelated inputs.

Perception. Perception concerns using audio–visual collaboration to infer a task-relevant state, signal, property, or structure from observations. Let $x_t = g(s_t)$ denote a task-relevant readout of the latent environmental state, where $g(\cdot)$ may extract cross-modal correspondence, temporal alignment, source location, object mask, source trajectory, separated sound, event category, spatial layout, acoustic property, or other perceptual descriptors. Perception is defined as

$$p_\theta\left(x_t \mid h_t, q\right), \tag{4}$$

with

$$\hat{x}_t = \arg\max_x p_\theta\left(x \mid h_t, q\right). \tag{5}$$

Here, $\hat{x}_t$ denotes the model-estimated perceptual variable at time step t. Its output is primarily an estimate or reconstruction of the current audio–visual content or environmental state. This category includes correspondence and synchronization, localization and grounding, segmentation and tracking, sound separation and enhancement, event and action recognition, scene parsing, and spatial or acoustic scene perception. These tasks need not already be deployed in an embodied agent; they are included when their outputs or learned capabilities can support embodied localization, environmental understanding, robust sensing, or subsequent action selection.

Reasoning. Reasoning concerns integrating and transforming audio–visual information to derive conclusions that cannot be obtained through direct recognition alone. Given a task query $q \in \mathcal{Q}$, it is defined as

$$p_\theta\left(y_t^R, r_t \mid h_t, q\right), \tag{6}$$

with

$$\left(\hat{y}_t^R, \hat{r}_t\right) = \arg\max_{y,r} p_\theta\left(y, r \mid h_t, q\right). \tag{7}$$

Here, y_t^R denotes the task-level reasoning output, which may take the form of a relational judgment, answer, explanation, physical conclusion, future prediction, counterfactual result, subgoal, or action plan, while r_t denotes an optional reasoning trace. Under this definition, Reasoning includes temporal and compositional reasoning, spatial and relational reasoning, causal and physical commonsense reasoning, and predictive or counterfactual reasoning. These tasks are included when the capabilities they evaluate can support embodied scene understanding, prediction, planning, or decision-making, even if their outputs are not yet executed by an embodied agent. The key distinction from Perception is that Reasoning composes multiple pieces of audio–visual information rather than directly estimating the current state.

Interaction. Interaction concerns selecting executable actions or responses using audio–visual observations. Under full observability, it is defined as

$$\pi_\theta\left(u_t \mid s_t, \ell\right). \tag{8}$$

TABLE III
REPRESENTATIVE DATASETS AND BENCHMARKS FOR AUDIO–VISUAL EMBODIED INTELLIGENCE. MORE BENCHMARKS SEE APPENDIX.

Benchmark	Year	Scale	Annotation	Task
Perception				
AudioSet [14]	2017	2.08M clips; 527 classes	Multi-label clip tags	Event recognition
MUSIC [15]	2018	714 videos; 11 instruments	Class labels + source tracks	Source separation
AVE [16]	2018	4,143 videos; 28 classes	Segment labels + boundaries	Event localization
VGGSound [17]	2020	200K+ clips (10 s)	Clip labels	Event recognition
LLP [18]	2020	11,849 videos; 25 classes	Video tags + temporal labels	AV video parsing
EPIC-KITCHENS-100 [19]	2020	100 h; ∼90K actions	Verb/noun labels + time spans	Action recognition / anticipation
VGG-SS [20]	2021	5,158 clips; 220 classes	Bounding boxes	Source localization
AVSBench (S4/MS3) [21]	2022	5,356 videos; 12,972 frames	Binary masks	AV segmentation
Reasoning				
Pano-AVQA [22]	2021	5.4K clips; 51.7K QA	QA + grounding boxes	Spatial AVQA
MUSIC-AVQA [23]	2022	∼9.3K videos; 45,867 QA	QA pairs	Spatio-temporal AVQA
PACS [24]	2022	1,526 videos; 13,400 QA	QA + object choices	Physical reasoning
Interaction				
SoundSpaces [25]	2019	103 scenes; 17.6M RIRs	Poses + geometry + goals	Audio-goal navigation
RH20T [26]	2023	110K+ sequences; 147 tasks	Trajectories + robot states	Imitation learning
ManiWAV [27]	2024	1,014 demos†; 4 tasks	AV trajectories	AV manipulation

Notes. Year denotes first public introduction, including preprints. Statistics follow the cited versions; MUSIC and AVSBench use their original benchmark versions. AV: audio–visual; QA: question–answer pairs; RIR: room impulse response. †ManiWAV's total sums the reported task-wise demonstration counts.

Under partial observability, it becomes

$$\pi_\theta\left(u_t \mid h_t, \ell\right). \tag{9}$$

Here, u_t may denote low-level motor commands or immediately executable action parameterizations, such as navigation waypoints, planned trajectories, sensor movements, manipulation actions, spoken responses, or collaborative signals. Thus, Interaction includes audio–visual navigation, active perception and exploration, manipulation with audio–visual feedback, situated dialogue, human–robot interaction, and general-purpose embodied task execution. The defining property of this category is that audio–visual collaboration directly affects an output deployed in a physical, simulated, or communicative loop, thereby changing the environment, the agent state, or subsequent observations.

C. Benchmarks and Metrics

Table III summarizes representative datasets and benchmarks, with additional resources provided in the Appendix. Perception benchmarks support recognition, grounding, and separation through clip-level labels, temporal annotations, and spatial supervision. Reasoning benchmarks evaluate spatial, temporal, and physical inference through audio–visual question answering. Interaction resources provide simulated environments and multisensory demonstrations for navigation and manipulation.

Evaluation metrics depend on the task, including recognition and answer accuracy, localization and segmentation quality, source separation quality, and interaction success or efficiency. Recorded demonstrations support policy learning, while closed-loop performance requires evaluation through executed actions and environmental feedback.

D. Preliminary

Data Representation. As illustrated in Fig. 3(a,b), audio–visual observations admit different representations according to their spatial, temporal, and spectral characteristics. Visual inputs include individual RGB or grayscale frames, sparse frame sequences, and dense video clips, providing appearance and motion information at different temporal resolutions. Depth and RGB-D observations supply geometric measurements, while thermal images capture temperature-related information. Panoramic and multi-view observations extend spatial coverage, and optical flow and pose representations explicitly describe motion and body structure. Audio can be represented as raw waveforms, STFT spectrograms, or mel-filterbank features. MFCCs, cochleagrams, and CQT/chroma features provide alternative descriptions of cepstral, auditory, and pitch-related structure. Stereo, binaural, and microphone-array recordings additionally preserve spatial information through interchannel phase, level, and arrival-time differences. Representation choices therefore determine which semantic, temporal, and spatial cues remain available for subsequent learning.

Model Architectures. Fig. 3(c) summarizes front-end encoders suited to these representations. Visual branches typically employ 2D convolutional neural networks (CNNs) or vision Transformers for individual frames, and 3D CNNs or video Transformers for spatiotemporal modeling [28], [29]. Dedicated branches encode optical flow, depth, and skeletons, while multi-view and panoramic encoders account for cross-view correspondence and viewing geometry. Audio branches learn directly from waveform samples or process time-frequency features using CNNs, convolutional recurrent networks, and spectrogram Transformers [30], [29], [31]. Spa-

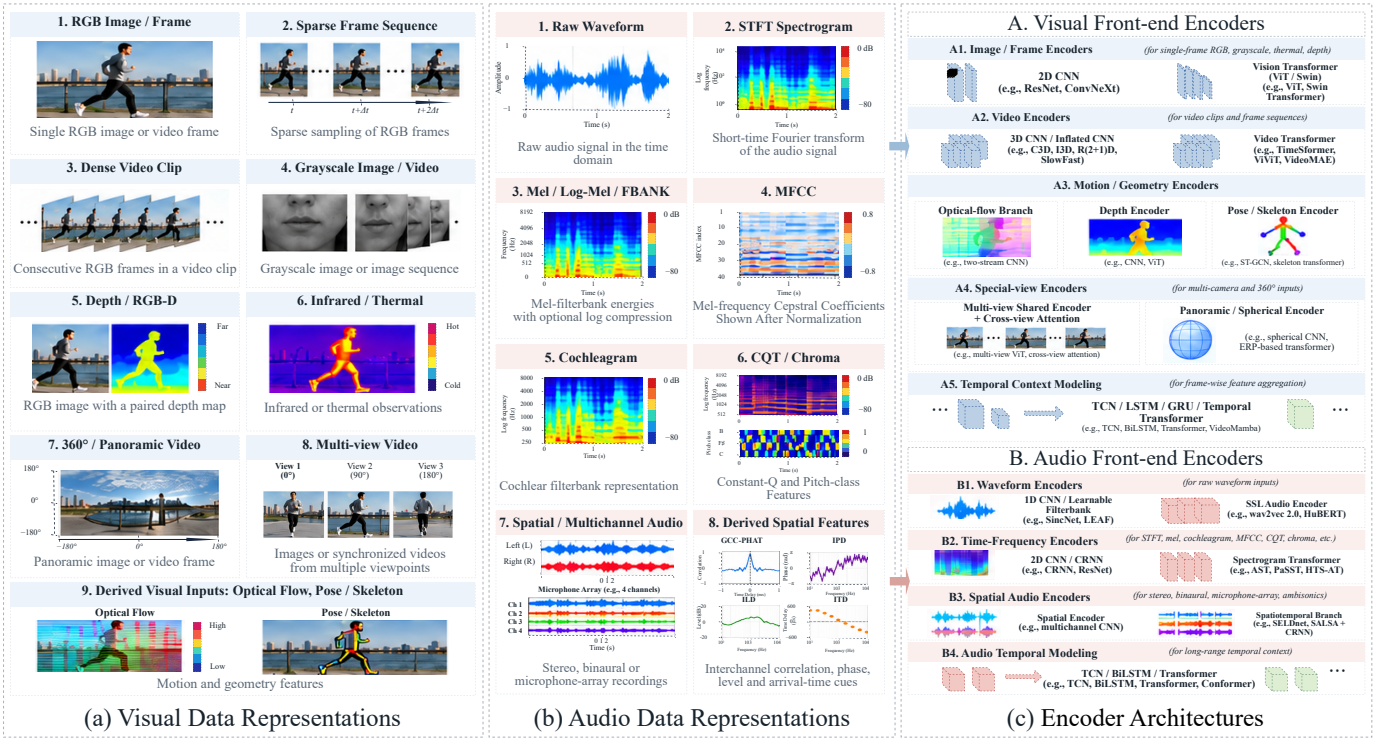


Fig. 3. Overview of data representations and front-end encoder architectures for audio–visual embodied intelligence. (a) Visual representations covering appearance, motion, geometry, and different viewing configurations. (b) Audio representations comprising raw waveforms and derived spectral, cepstral, pitch-related, and spatial features. (c) Representative visual and audio encoders with temporal context modeling modules for processing these inputs.

tial audio encoders further preserve interchannel relationships for directional understanding [32], [33]. Temporal convolution, recurrent networks, and attention modules aggregate context across visual frames and audio windows. The resulting features support downstream audio–visual alignment and fusion through shared embedding spaces or cross-modal attention [4], [34].

III. Perception

A. Consistency

Definition. Audio-visual consistency examines whether auditory and visual observations correspond in semantic content, timing, and spatial configuration. It encompasses three closely related task families. *Semantic correspondence* is exemplified by L^3-Net [35], which formulates audio-visual correspondence as distinguishing matched from mismatched audio–image pairs. *Temporal synchronization* concerns synchronization detection and temporal offset estimation: SyncNet [36] establishes an influential formulation for lip–speech alignment, while AVTS [37] extends synchronization-based self-supervision to general video. *Spatial alignment* assesses whether acoustic direction agrees with visual arrangement, exemplified by left–right correspondence discrimination in Telling Left from Right [32] and cross-view matching between panoramic vision and spatial audio in AVSA [33]. Together, these tasks establish whether modalities convey compatible evidence about what, when, and where, rather than explicitly locating the supporting pixels or event intervals, as in grounding [16].

Methods. Across these dimensions, methods progress from correspondence discrimination toward shared representation learning and robust alignment under realistic observations. Their supervision is often obtained by contrasting compatible observations with semantic mismatches, temporal shifts, or spatial transformations, while reconstruction objectives provide an alternative to explicit compatibility classification.

Semantic correspondence initially exploits naturally paired videos for cross-modal prediction, knowledge transfer, and matched–mismatched discrimination [38], [39], [30], [35], subsequently supporting transferable embeddings and multiple semantic associations [40], [41], [42]. Shared representations are learned through cross-modal clustering [43], [44], contrastive objectives [45], [46], [47], [48], [29], and masked learning [49], [31], [50]. Language further anchors broader multimodal semantic spaces [51], [4], [52], [53], while data curation and relaxed pairing assumptions address noisy co-occurrence and incomplete correspondence [54], [55], [56]. The central challenge is to distinguish genuine semantic compatibility from coincidental co-occurrence without treating semantically related samples as negatives.

Temporal synchronization develops from speech alignment and aligned–shifted discrimination [57], [36], [58], [37], [59] toward explicit offset estimation and fine-grained cross-modal interaction [60], [61], [62], [63], [64]. Synchronization constraints also support speech separation and lip generation [65],

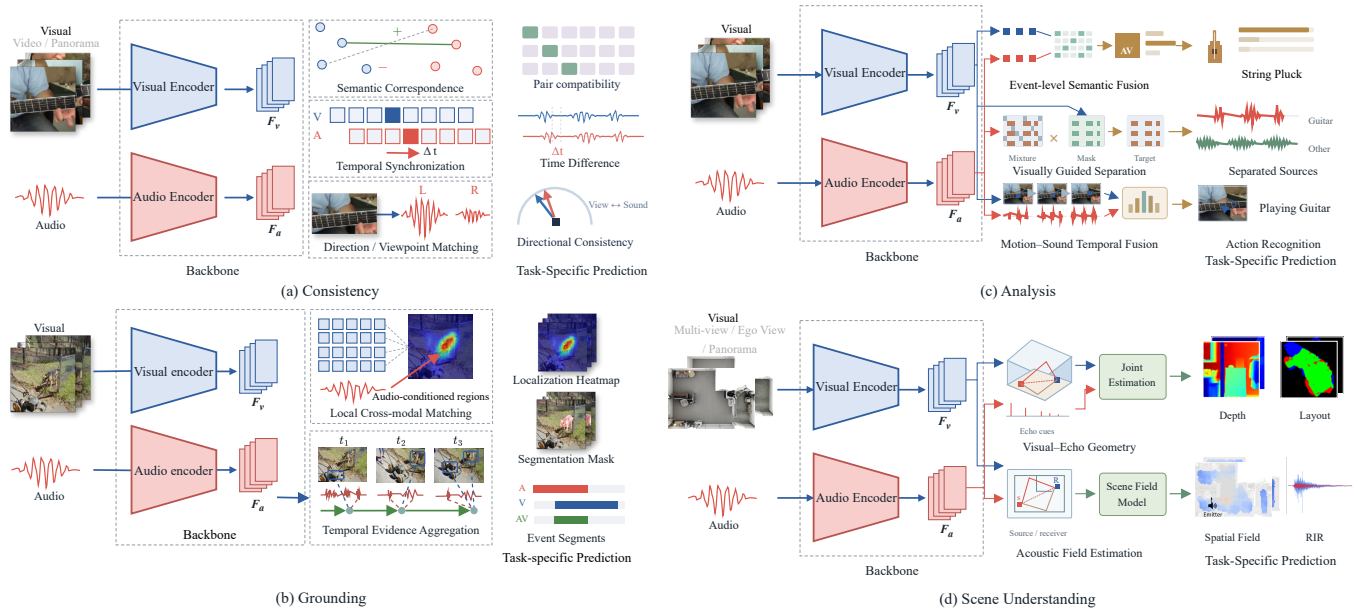


Fig. 4. Overview of perception.

[66], [67]. More realistic formulations address sparse temporal cues [68], [69], [70], [71], relaxed alignment and temporal augmentation [55], [72], [73], and synchronization-sensitive evaluation, streaming, manipulation detection, and generation [74], [75], [76], [77]. This progression replaces the assumption of dense, speech-like timing cues with models that recover correspondence from irregular and potentially corrupted observations.

Spatial alignment learns directional compatibility through channel swapping, viewpoint matching, and spatial contrastive supervision [32], [33], [78]. Complementary methods encode spatial correspondence through visually conditioned spatial-audio reconstruction [79], [80], extending to partial supervision, pseudo-binaural training, and localization or geometric constraints [81], [82], [83], [84]. Object-aware generation and spatially sensitive contrastive learning strengthen this connection [85], [86]. Egocentric reconstruction, motion supervision, and richer spatial recordings further shift alignment toward geometry that remains coherent across changing observer poses [87], [88], [89]. Here, binaural or ambisonic cues complement visual layout, but their interpretation must account for microphone configuration and the changing relationship between acoustic and visual coordinate frames.

Discussion. Consistency is conditional rather than synonymous with co-occurrence, exact simultaneity, or shared visibility. Silent objects, off-screen sources, intermittent events, recording delays, reverberation, and ego-motion can produce valid modality-specific evidence; conversely, dataset biases can create apparent agreement without genuine correspondence. For embodied agents, these dimensions must therefore be maintained jointly as actions alter scene content, event timing, and observer geometry. Navigation demonstrates the downstream value of cross-modal associations [90], [25], [91], [92], while interaction-based learning and active listening connect consistency to action and renewed observation [93], [94]. A promising direction is uncertainty-aware, action-conditioned consistency that preserves modality-specific evidence and actively verifies ambiguous associations.

B. Grounding

Definition. Audio-visual grounding identifies *where* and *when* relevant evidence is instantiated in multimodal observations. It encompasses two complementary task families. *Pixel grounding* associates sound with visible regions, from sound-source localization exemplified by Senocak *et al.* [95] to audio-visual segmentation (AVS), which delineates sounding objects with pixel-wise masks [21]. *Temporal grounding* identifies event-relevant segments or boundaries: Audio-Visual Event Localization (AVEL) localizes jointly audible and visible events [16], while Audio-Visual Video Parsing (AVVP) extends this formulation to audio-only, visual-only, and joint events [18]. Unlike consistency, which assesses cross-modal compatibility, grounding explicitly recovers spatial or temporal support; unlike event recognition, its objective extends beyond predicting event categories.

Methods. Existing approaches progress from coarse source localization and segment classification toward dense, instance-aware, and open-world grounding.

Sound-source localization learns audio-conditioned response maps through hard-negative discrimination and local cross-modal alignment [20], [96]. Object proposals and correspondence-derived pseudo labels extend localization toward object-level detection [97], [98], while category-wise decomposition and explicit handling of silent objects, off-screen sounds, and mismatched pairs improve source disambiguation [99], [56]. Language guidance supplies source-specific semantic priors [100], [101]; slot-based representations and joint speech–sound grounding further address heterogeneous or overlapping sources [102], [103].

Dense audio-visual segmentation replaces coarse regions with object masks. Building on pixel-wise audio-visual interaction [21], methods improve segmentation through shared–specific representation disentanglement, neighboring-frame supervision, visually guided acoustic retrieval, and source-specific soundtrack decomposition [104], [105], [106], [107]. Subsequent work addresses long-video spatiotemporal grounding, visual-prior bias, referring expressions, and instance tracking [108], [109], [110], [111]. Text-mediated prompting and omnimodal conditioning further connect acoustic evidence with segmentation foundation models and richer target specifications [112], [113], extending the objective from identifying sounding regions to selecting and maintaining the relevant instance.

Temporal event grounding initially uses cross-modal attention and temporal aggregation for segment-level localization [114], [115], [116], with background suppression and structured temporal modeling improving event discrimination [117], [118]. Under weak supervision, modality-specific clues, label denoising, and selective fusion reduce ambiguity in video-level labels [119], [120], [121]; segment recovery, prototype-based pseudo labels, and uncertainty-aware or long-range modeling strengthen temporal predictions [122], [123], [124], [125]. Recent approaches extend grounding to unlabeled observations, dense events in untrimmed videos, and unseen categories [126], [127], [128], [129], while textual semantics and pretrained representations provide additional contextual guidance [130], [131].

Discussion. Spatial and temporal grounding share the challenge of selecting evidence that actually supports an event, rather than exploiting visual saliency, category priors, or recording-level co-occurrence. Coarse maps and segments are efficient but can merge sources or obscure boundaries; dense masks and intervals provide finer support while increasing supervision demands. For embodied agents, grounding must remain reliable under ego-motion, intermittent sound, occlusion, and changing source activity, including cases where no visible region supports an audible event. A promising direction is streaming, uncertainty-aware spatiotemporal grounding that maintains source identities and event boundaries while distinguishing joint evidence from modality-specific observations.

C. Analysis

Definition. Audio-visual analysis recognizes semantic content and decomposes acoustic mixtures through complementary sensory evidence. It encompasses *event recognition*, supported by transferable acoustic representations exemplified by SoundNet [30]; *source separation*, which reconstructs individual signals using visual cues, as in The Sound of Pixels [15] and Looking to Listen [132]; and *action recognition*, which identifies agent-centered behaviors through motion and sound, exemplified by EPIC-Fusion [133]. These tasks predict categories or recover source signals, complementing grounding's spatial and temporal support estimation without requiring explicit question answering or action planning.

Methods. *Event recognition* advances from attentive fusion and robustness analysis [134], [135], [136] to balanced multimodal optimization, exemplified by Gradient Blending [137], [138], [139]. Multimodal Bottleneck Transformer (MBT) and shared architectures enable efficient cross-modal interaction [34], [140], [141], while clustering, masked reconstruction, and contrastive learning yield transferable representations [43], [49], [31], [50]. Language-guided supervision and large-scale correspondence learning extend transfer toward broader event vocabularies [142], [53].

Source separation progresses from object-conditioned decomposition and Mix-and-Separate training to learning from multi-source videos [143], [15], [144]. Motion, gesture, grounding, and multi-resolution cues refine source association [145], [146], [147], [148], [149], [150]; speech-oriented methods exploit attention, facial identity, and lip motion, notably VisualVoice [151], [152], [153]. Query-based separation, off-screen source modeling, and binaural spatial cues relax category and visibility assumptions [154], [155], [156]. Test-time adaptation and continual learning address unfamiliar sources [157], [158], while diffusion, adaptive fusion, and speaker-count estimation broaden reconstruction and deployment capabilities [159], [160], [161], [162].

Action recognition combines temporal binding and auditory–visual pathways [133], [28], [163] with uncertainty weighting and selective computation, including Listen to Look and AdaMML [164], [165], [166]. Contrastive learning and SoundingActions strengthen action-related representations [167], [168]; TIM, Sound Bridge, and cross-attention experts model asynchronous intervals, viewpoint changes, and complementary sensory structure [169], [170], [171]. Domain alignment, audio-guided adaptation, incremental learning, and multi-objective optimization improve generalization [172], [173], [174], [175], while EgoAdapt selectively processes informative inputs under computational constraints [176].

Across recognition tasks, data resources expand from AudioSet and VGGSound to modality-aware event annotations [14], [17], [177], and from kitchen activities to audible interactions, detailed household activities, and industrial assembly [19], [178], [179], [180].

Discussion. These tasks share a central difficulty: useful cross-modal evidence must be distinguished from incidental co-occurrence. Modality dominance, silent actions, off-screen sources, reverberation, and incomplete labels can undermine recognition and separation. Moreover, improved clip classification or synthetic-mixture reconstruction does not establish reliable embodied perception. A promising direction is streaming, uncertainty-aware analysis that maintains event, action, and source hypotheses under ego-motion, changing visibility, unknown source counts, and emerging categories. Coupling separation with recognition may improve source attribution, but must avoid amplifying reconstruction errors or semantic priors unsupported by observations.

D. Scene Understanding

Definition. Audio-visual scene understanding recovers environmental structure and acoustic properties through visual observations and sound propagation cues. *Geometric estimation* infers depth, layout, or articulated geometry, exemplified by echo-constrained single-view reconstruction [181]

and VisualEchoes [182]. *Acoustic field estimation* predicts how environments transform sound, from image-conditioned room impulse responses (RIRs) in Image2Reverb [183] to source–listener-dependent responses in Neural Acoustic Fields (NAF) [184]. The former recovers spatial structure; the latter models propagation conditioned on geometry, materials, and sensing configurations.

Methods. *Geometric estimation* progresses from echo-based depth prediction, exemplified by BatVision [185], to joint visual–acoustic estimation and material-aware fusion in VisualEchoes and Beyond Image to Depth [182], [186]. Richer echo representations and scale-aware modeling improve geometric recovery [187], [188], [189]. Audio-Visual Floorplan Reconstruction extends inference beyond visible surfaces [190], while active mapping and acoustic SLAM exploit reflections during exploration [191], [192]. PoseKernelLifter combines images and body-induced acoustic responses for metric pose estimation [193]; complementary acoustic-only studies address position variation, non-stationary excitation, and multiple people [194], [195], [196], [197], demonstrating geometric information available beyond direct visual observation.

Acoustic response and field modeling develops from scene-aware rendering and visual acoustic matching [198], [199] toward joint audio-visual RIR estimation and material-conditioned synthesis [200], [201], [202]. NAF, INRAS, and contextual neural fields represent propagation across source–listener configurations [184], [203], [204]. AV-NeRF, AV-GS, and NeRAF connect acoustic prediction with visual scene representations [205], [206], [207], while AV-DAR incorporates differentiable acoustic tracing [208]. Sparse-observation methods reconstruct responses or synthesize novel-view audio from limited measurements [209], [210], [211], [212], [213]; occlusion-aware fields, active sampling, and one-shot scene construction further expand reconstruction settings [214], [215], [216].

Discussion. Vision supplies dense appearance and boundaries, whereas sound offers indirect distance and propagation constraints. Their joint interpretation remains ambiguous because geometry, materials, multipath, and source–receiver configuration all affect acoustic observations. Audio-Visual BatVision and Real Acoustic Fields support evaluation beyond simulation [217], [218], but controlled excitation, calibration requirements, sparse measurements, and static-scene assumptions still limit deployment. For embodied agents, a promising direction is to jointly update geometric and acoustic representations from streaming observations, combining passive listening with selective active sensing. Such systems should distinguish measured evidence from plausible synthesis, track uncertainty as objects and observers move, and use predicted propagation to guide informative observations without conflating acoustic responses with uniquely determined geometry.

E. Discussion

Table IV compares representative perception methods in terms of their core mechanisms, supervision, outputs, and key settings. The comparison highlights how audio–visual learning supports shared representations, spatial and temporal grounding, event and source analysis, and scene modeling.

IV. Reasoning

Audio-visual reasoning integrates sensory evidence to derive relations, explanations, predictions, or plans that extend beyond direct recognition. Sound can reveal hidden sources, interaction dynamics, and temporally preceding events, while vision constrains their entities and spatial context. We organize this section into *Specialized-Model Reasoning*, *Foundation-Model Reasoning*, and *Agentic Reasoning*, according to how inference is structured. These methodological families overlap: a foundation model can participate in a specialized spatial pipeline or an agentic workflow. Their outputs support embodied decisions; executing actions and incorporating environmental feedback are treated in Section V.

A. Specialized-Model Reasoning

Definition. Specialized-model reasoning introduces task-specific representations, fusion mechanisms, or inference structures for well-defined audio-visual problems. It encompasses *cross-modal event reasoning*, which composes temporal, counting, and comparative relations, exemplified by MUSIC-AVQA [23]; *situated spatial reasoning*, which interprets source–object relations relative to an observer or reference frame, exemplified by Pano-AVQA [22]; and *physical-causal reasoning*, which infers affordances and interaction consequences from appearance and sound, exemplified by PACS [24]. Specialization concerns the reasoning formulation rather than model size: task-specific pipelines may also employ pretrained foundation models.

Methods. Across these directions, methods develop from global feature fusion toward explicit evidence selection, structured relations, and robustness to sensory or dataset shortcuts.

Cross-Modal Event Reasoning connects sounds, objects, and actions to answer questions requiring their joint interpretation. MUSIC-AVQA combines spatial source association with question-conditioned temporal aggregation, while AVQA broadens coverage from musical performances to everyday events [23], [219]. Subsequent methods improve which evidence enters the inference: PSTP-Net progressively selects relevant temporal and spatial content, CAD aligns spatial, temporal, and semantic information, and QA-TIGER uses question-aware Gaussian experts to represent both consecutive and separated temporal evidence [220], [221], [222]. Parameter-efficient adaptation offers a complementary route: LAViSH transfers frozen visual backbones through latent audio-visual adapters, and DG-SCT uses cross-modal prompts across spatial, channel, and temporal dimensions [223], [224]. These approaches improve the representations supporting relational answers without requiring a new backbone for every task. Balanced MUSIC-AVQA evaluation and multimodal collaborative debiasing expose reliance on answer frequencies and linguistic shortcuts [225], [226]; AVQACL extends the problem to sequentially learned tasks, where preserving previously acquired spatiotemporal knowledge becomes essential [227].

TABLE IV
REPRESENTATIVE METHODS FOR AUDIO–VISUAL PERCEPTION.

Method / task	Core mechanism	Supervision	Output	Key setting
Consistency				
AVTS [37] *Temporal sync.*	Contrast aligned and time-shifted AV clips	Self-sup. Temporal pairing	AV features + sync scores	Temporally paired clips
TLR [32] *Spatial alignment*	Detect stereo-channel flips from visual source cues	Self-sup. Channel flips	Spatial features + flip scores	Stereo audio
CAV-MAE [31] *Semantic alignment*	Cross-modal contrast + masked reconstruction	Self-sup. Paired clips	AV embeddings	Representation pretraining
Grounding				
LVS [20] *Source localization*	Local AV similarity + hard-negative mining	Self-sup. AV pairing	Source heatmaps	Image-plane localization
AVS baseline [21] *AV segmentation*	Temporal pixel-wise AV interaction (TPAVI)	Sparse / dense pixel masks	Sounding-object masks	Single / multiple sounding sources
HAN [18] *Temporal parsing*	Hybrid temporal attention + multimodal MIL pooling	Weak video tags	Modality-wise event segments	No temporal labels for training
EEMC [110] *Referring segmentation*	AV-enhanced expressions + temporal memory for mask prompting	Masks + referring expressions	Referred-object masks	Text-specified targets
Analysis				
MBT [34] *Event recognition*	AV information exchange through bottleneck tokens	Event labels	Event classes	Clip-level recognition
Sound of Pixels [15] *Source separation*	Visually conditioned spectrogram masking	Self-sup. Synthetic mixtures	Separated audio	Mix-and-Separate training
EPIC-Fusion [133] *Action recognition*	RGB–flow–audio fusion before temporal aggregation	Verb / noun labels	Action classes	Egocentric video
Scene Understanding				
VisualEchoes [182] *Spatial pretraining*	RGB–echo relative-orientation prediction	Self-sup. Echo orientation	Spatial image features	Echo pretraining; RGB-only testing
NeRAF [207] *Acoustic field estimation*	Acoustic fields conditioned on radiance-field scene features	RGB + RIR targets	Novel-view RGB + spatialized RIRs	Scene-specific fitting

Notes. AV: audio–visual; Self-sup.: self-supervised; MIL: multiple-instance learning; RIR: room impulse response. TLR: *Telling Left from Right*. EEMC is the Ref-AVS baseline. Supervision describes the listed learning stage, excluding off-the-shelf encoder pretraining. Key settings summarize the original formulations; groups are not exclusive.

The resulting progression is from combining modalities to selecting, relating, and retaining question-relevant evidence.

Situated Spatial Reasoning develops from panoramic source–object questions toward dynamic, viewpoint-dependent inference [22]. Audio-visual floorplan reconstruction and Chat2Map provide supporting spatial representations, although their map outputs belong primarily to scene perception [190], [228]. Building on explicit geometry, SAVVY aggregates egocentric audio-visual tracks into a global dynamic map and transforms them to the queried viewpoint [229]. Hear You Are couples vision with spatial sound for source association and spatial questions, while JAEGER combines RGB-D observations and Ambisonics to support joint three-dimensional grounding and reasoning [230], [231]. SceneBind supplies complementary object-centered semantic–spatial representations [232]. Fine-grained temporal grounding in R-AVST and trajectory reasoning in ST-Omni-R1 further address changing source identity, location, and visibility [233], [234]; EgoSound extends evaluation to egocentric sound understanding [235]. Unlike estimating a sound direction alone, the central inference is how acoustic and visual entities relate across time and coordinate frames, particularly when a source becomes occluded or the observer moves.

Physical-Causal Reasoning links audible interactions and visible object behavior to latent properties and possible outcomes. Visually indicated sound prediction establishes an early connection between contact appearance and acoustic response [38], while PACS explicitly asks which object satisfies a physical or functional requirement [24]. Disentangled Counterfactual Learning separates static object factors from interaction dynamics; CF-PACR reduces direct visual shortcuts; and robust disentangled learning reconstructs missing modality information through shared physical semantics [236], [237], [238]. These approaches develop from learning correlations toward controlling which factors support a physical conclusion. Related visual benchmarks, including PhysBench, CausalVQA, and minimal video pairs, offer diagnostics for physical knowledge, counterfactual questions, and shortcut sensitivity, but do not independently establish the contribution of sound [239], [240], [241]. Conversely, PAVAS conditions audio synthesis on estimated physical variables, and FlatSounds tests whether changing one physical factor induces the expected acoustic change [242], [243]. Such generative probes complement question answering by examining pre-

dicted consequences; latent interventions and plausible sounds nevertheless remain weaker evidence than physically verified causal mechanisms.

Discussion. Specialized models make event relations, coordinate transformations, and physical factors explicit, supporting targeted analysis of inference failures. Their limitations arise from restricted question vocabularies, dataset-specific associations, and assumptions about source visibility or acoustic calibration. Answer accuracy should therefore be paired with evidence grounding, spatial consistency, and controlled changes to the relevant observations [225], [229], [243]. For embodied agents, a promising direction is to maintain uncertain relational and physical hypotheses as viewpoints and observations change, distinguishing observed properties from inferred consequences and revising conclusions when new sound or visual evidence contradicts them.

B. Foundation-Model Reasoning

Definition. Foundation-model reasoning integrates diverse audio-visual tasks within a reusable language-conditioned model. *Native omnimodal reasoning* directly incorporates audio and visual representations into a unified inference system, exemplified by video-SALMONN [244]; *multi-step reasoning* learns or generates intermediate inference states, exemplified by video-SALMONN-o1 [245]; and *long-context reasoning* links evidence across extended observations, supported by temporally annotated resources such as LongVALE [246]. Native input permits modality-specific encoders but differs from replacing all auditory evidence with transcripts; long-context reasoning requires connecting distant evidence rather than simply accepting more frames.

Methods. The main trajectory is from attaching sensory representations to language models toward coordinated modality alignment, evidence-grounded deliberation, and efficient preservation of temporal context.

Native Omnimodal Reasoning initially connects pretrained sensory encoders to language backbones. ImageBind supplies a shared embedding space, PandaGPT transfers it to multimodal instruction following, and Video-LLaMA introduces audio and video branches for language-conditioned understanding [4], [247], [248]. OneLLM explores unified modality alignment, while VideoLLaMA 2 strengthens spatiotemporal modeling and audio integration [249], [250]. More coordinated fusion appears in video-SALMONN, which represents speech, environmental sound, and visual content through a multi-resolution connector [244]. CREMA uses modular adaptation and fusion to incorporate additional sensory information efficiently; VITA and Ola develop multimodal instruction training and progressive alignment [251], [252], [253]. Qwen2.5-Omni introduces temporally aligned audio-video processing, and Qwen3-Omni extends unified understanding through a mixture-of-experts Thinker–Talker architecture [254], [255]. Across these designs, broad modality coverage is an enabling interface rather than proof of joint inference. OmniBench, WorldSense, AV-Odyssey, and Daily-Omni test complementary evidence use and expose the need to distinguish perception failures from reasoning failures [256], [257], [258], [259].

Multi-Step Reasoning advances from generating explanatory text toward supervising consequential intermediate decisions. video-SALMONN-o1 couples reasoning-intensive audiovisual data with process-level preference optimization, comparing alternative continuations to improve the reasoning trajectory [245]. AURELIA provides a complementary inference-time approach, using actor–critic feedback to refine structured reasoning without additional parameter training [260]. Reinforcement learning extends this direction: AVATAR reuses trajectories and weights planning and synthesis stages, while OmniVideo-R1 strengthens query grounding and modality-sensitive fusion [261], [262]. LatentOmni interleaves textual steps with audio-visual latent states, addressing information lost when continuous sensory evidence is compressed into words [263]. Related preference learning in AVERE targets unsupported cue–emotion associations and hallucinated audio-visual evidence, illustrating that optimization must constrain sensory faithfulness as well as answer plausibility [264]. RivaBench and AVReasonBench supply reasoning-focused evaluation [245], [260]; however, longer explanations are not necessarily better inference. Each decisive step should remain attributable to the observation that supports it, and improvements should be assessed at comparable inference budgets.

Long-Context Reasoning must preserve event identity and dependencies while selecting information from temporally dense inputs. Video-MME evaluates videos across durations and input settings, providing broader context for long-video assessment [265]. LongVALE supplies multimodal event boundaries and correlated descriptions, and TriSense uses query-conditioned modality weighting to identify relevant audio-visual-speech moments [246], [266]. These event-centered formulations reduce the mismatch between global video summaries and questions about specific temporal relationships. Streaming methods additionally require bounded state: video-SALMONN S consolidates observations through test-time memory updates and a modality-aware reader, while OmniMem allocates separate audio and visual memory budgets and selects informative cache states [267], [268]. LVOmniBench specifically extends evaluation to long audio-video understanding [269]. Here, compression and recurrent state support inference within an observation stream; independently addressable stores retrieved through agent decisions are discussed under memory augmentation. The unresolved tradeoff is retaining brief acoustic events and distant dependencies without making computation grow with every observed frame.

Discussion. Foundation models broaden task coverage and enable shared reasoning representations, but common language priors can override weak acoustic evidence, and temporal compression can discard the events needed for a correct conclusion. Modality ablations, balanced questions, and evidence-linked explanations are therefore more informative than aggregate accuracy alone [257], [259], [260]. For embodied use, a promising direction is adaptive inference that combines rapid sensory updates with selective multi-step deliberation, preserves timestamps and uncertainty during compression, and revises earlier conclusions when new observations arrive. Evaluation should jointly report reasoning correctness, evidence retention, memory cost, and latency under causal access to

the stream.

C. Agentic Reasoning

Definition. Agentic reasoning organizes inference as a controllable loop whose next operation depends on intermediate observations. It encompasses *tool-augmented reasoning*, which selects external operations to acquire or verify evidence; *memory-augmented reasoning*, which writes and retrieves addressable records of past experience; and *multi-agent collaboration*, which distributes inference across cooperating software agents. Representative systems include OmniAgent, EgoLife's EgoRAG, and MAGNET, respectively [270], [271], [272]. These operations concern information processing; robot motion, manipulation, and executed communication belong to Interaction.

Methods. Methods progress from predefined analysis pipelines to selective evidence acquisition, persistent experience retrieval, and coordinated inference over distributed observations.

Tool-Augmented Reasoning uses external perception or retrieval modules to resolve uncertainty in a question. AVAgent introduces an audio-visual planning and verification workflow for representation alignment, serving as a data-centric precursor rather than an embodied decision system [273]. Video-RAG extracts auxiliary speech, text, and object information through external tools, while long-form retrieval and AVQA pipelines add segment selection and answer self-feedback [274], [275]. These approaches establish useful components, although fixed retrieval alone does not constitute an adaptive agent. OmniAgent subsequently uses sound to identify promising intervals and orchestrates more detailed analysis around them [270]. OmniRAG-Agent combines image–audio retrieval banks with multi-turn planning and policy optimization for tool use and answering [276]. A separate, native active-perception approach learns iterative observation–thought–action cycles and selectively records acquired evidence [277]. The methodological shift is toward deciding what to inspect next based on missing evidence, rather than repeatedly processing the entire recording. Tool provenance and temporal coordinates must be preserved when intermediate outputs are converted into text.

Memory-Augmented Reasoning separates experience storage from the model's immediate context. VideoAgent provides a related visual foundation by storing temporal event descriptions and object-tracking states that an agent can query [278]. EgoLife extends memory-oriented assistance to prolonged egocentric audio-visual experience: EgoRAG organizes timestamped descriptions hierarchically and retrieves relevant past observations for answering [271]. HippoMM further connects audiovisual episode formation, memory consolidation, and cross-modal associative retrieval [279]. Retrieval can also compensate for incomplete inputs, as studied by MissRAG, but retrieved external examples are contextual evidence rather than observations of the current episode [280]. EgoMemReason probes entity changes, event order, and recurring behavior over long horizons, offering a complementary memory diagnostic whose visual focus does not by itself demonstrate acoustic reasoning [281]. These developments motivate memories that retain source identity, observation time, and links to original clips alongside summaries. Otherwise, a fluent summary can erase a brief alarm or contact sound, and retrieving an obsolete state can produce a confident but incorrect belief about the present environment.

Multi-Agent Collaboration distributes reasoning by evidence source or computational role. MAGNET retrieves relevant videos, instantiates audio-visual agents to derive local evidence, and uses a meta-agent to assemble grounded, stepwise answers across the collection [272]. Its contribution is the coordination of evidence that exceeds a single context, rather than merely generating several answers to the same input. XGC-AVis adopts a complementary division into perception, planning, execution, and reflection, using independent proposals and arbitration to improve temporal selection and audiovisual understanding [282]. Actor–critic refinement in AURELIA provides a related role-based verification mechanism, linking collaborative critique to multi-step inference [260]. AVHaystacks and XGC-AVQuiz expose cross-video and temporal-alignment challenges [272], [282]. Across these systems, useful collaboration requires agents to exchange the supporting intervals and modality-specific observations, not only conclusions. Independent roles can reveal missing evidence, but shared backbones or captions can also propagate the same mistake through every stage.

Discussion. Agentic systems make evidence acquisition and inference resources explicit, enabling selective computation and reuse of past experience. Their flexibility introduces dependence on tool reliability, memory fidelity, and coordination quality. Comparisons should therefore control model access, token budgets, and tool calls, and distinguish improved evidence coverage from additional computation. For embodied agents, a promising direction is a persistent reasoning loop that detects conflicting observations, revisits their original audio-visual support, and updates beliefs before proposing an action. Progress should be measured through grounded answers, recovery from incorrect tool or memory outputs, and the usefulness of inferred beliefs for subsequent closed-loop behavior.

D. Discussion

The three families offer complementary ways to preserve the connection between evidence and conclusion: specialized models encode task structure, foundation models share capabilities, and agentic systems control access to information. Their common limitation is that offline answer quality does not establish reliable embodied decisions. Evaluation should test whether removing or changing relevant audio alters the conclusion appropriately, whether spatial and temporal evidence remains consistent, and whether additional reasoning justifies its computational cost. Connecting these diagnostics to later action outcomes would clarify when audio-visual reasoning supplies information that an embodied agent can actually use.

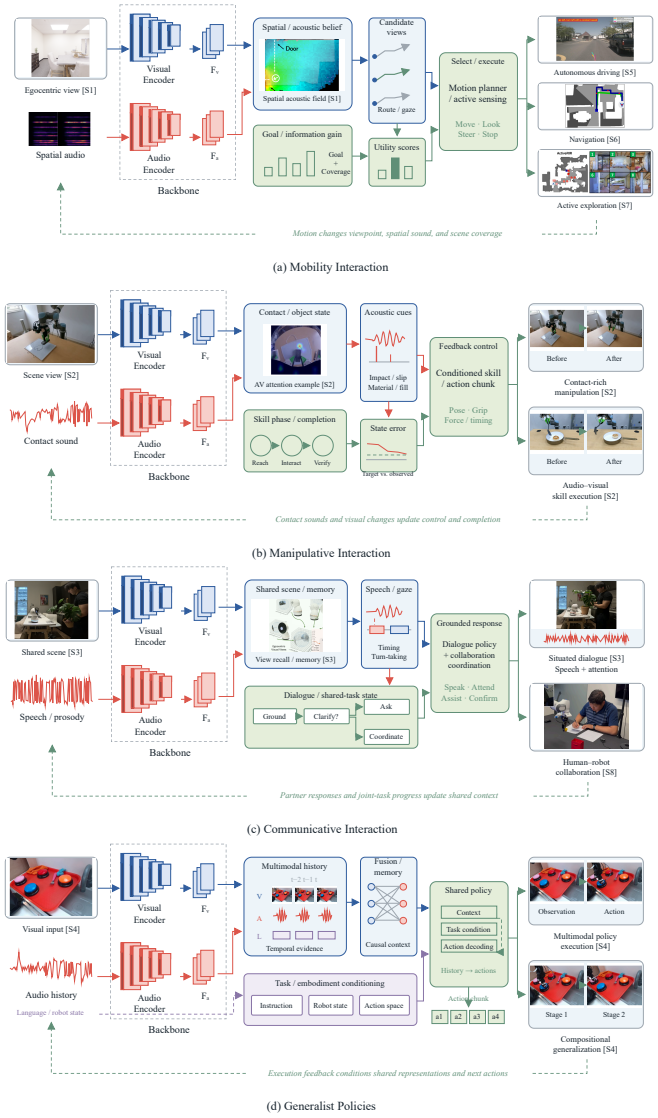


Fig. 5. Caption

V. Interaction

A. Mobility Interaction

Audio-visual mobility interaction concerns embodied settings in which multimodal observations directly influence how an agent moves through the physical environment. Unlike spatial perception or scene understanding, which estimate where objects, surfaces, or sound sources are located, mobility interaction closes the perception–action loop by converting sensory evidence into navigation decisions, exploratory motions, trajectories, or low-level locomotion commands. Auditory sensing is particularly complementary to vision because it provides omnidirectional evidence about targets and events outside the current field of view and can remain informative under visual occlusion, while vision supplies dense geometric and semantic information for route planning and collision avoidance. Depending on the purpose of motion, we organize existing work into three task families: *navigation*, where movement is used to reach a target; *active exploration*, where actions are selected primarily to acquire more informative observations; and *autonomous driving*, where multimodal sensing contributes to safety-critical vehicle understanding, planning, and control.

Audio-visual navigation (AVN) is the most established form of mobility interaction, where an embodied agent must reach a sounding or semantically associated target from streams of egocentric visual and auditory observations. Early works such as Look, Listen, and Act [90] and SoundSpaces [25] establish this closed-loop setting through learned mapping and reinforcement-learning policies in realistic 3D environments. Subsequent approaches introduce dynamically predicted waypoints and acoustic memory [91], semantic grounding and persistent multimodal memory for intermittent sounds [92], and robustness to acoustic interference or multiple goals [283], [284]. Navigation is further extended with language-based assistance and richer information gathering through AVLEN, ORAN, and CAVEN [285], [286], [287]. More recent approaches increasingly emphasize generalization and realistic sensory conditions: RILA [288] uses multimodal foundation models and language-based planning for zero-shot semantic AV navigation, BeDAViN/ENMuS^3 [289] addresses multiple simultaneous sound sources, and auxiliary action-description learning improves both navigation and interpretability [290]. SAVN-CE [291] further removes the discrete-grid assumption and enables fine-grained movement with spatially and temporally continuous audio, moving AV navigation toward more realistic embodied environments.

In active exploration, motion is not primarily a means of reaching a predefined destination but an instrument for improving perception or reducing uncertainty. Move2Hear [94] learns movement policies that reposition an agent's camera and microphones to improve audio-visual source separation, while Finding Fallen Objects [292] requires an agent to actively search for an object by integrating an earlier impact sound with subsequent visual observations. Active acoustic sensing has also been coupled with mapping: Learning to Map Efficiently by Active Echolocation [191] jointly determines navigation actions and when probing sounds should be emitted to reveal geometry beyond visible regions. ActiveRIR [293] extends this principle to acoustic environment modeling, selecting movement and measurement locations according to information gain. These works establish a key property of embodied audio-visual intelligence: sensory evidence is action-dependent, and an agent can deliberately change its pose or sensing configuration to obtain information unavailable from passive observation.

Audio provides a complementary sensing channel for autonomous vehicles because safety-critical events such as sirens, horns, collisions, and approaching vehicles may become audible before they are reliably visible. Early studies therefore incorporate acoustic cues into traffic-scene understanding and emergency-vehicle awareness [294], [295], while recent work further explores environmental sound sensing, localization, and joint audio-visual traffic understanding [296], [297], [298]. More recently, omnimodal foundation models are beginning to connect audition with driving-oriented understanding and action. EchoVLA [299], built upon the

audio-capable Qwen2.5-Omni backbone [254], couples camera streams with online spoken instructions and directly predicts driving trajectories, allowing both linguistic content and paralinguistic cues to influence vehicle behavior. Newer omni backbones further unify visual, acoustic, and linguistic perception [255], [300], while NVIDIA Cosmos 3 [301] jointly represents language, vision, sound, and action within an omnimodal world-model framework and supports autonomous-vehicle action modeling. Together, these developments indicate a transition from audio-assisted traffic perception toward omnimodal world-action models in which environmental sounds, human speech, physical-world representations, and vehicle actions can increasingly be modeled within a shared framework.

Discussion. Mobility interaction demonstrates how audio-visual intelligence changes once perception becomes embedded within a closed action loop. Navigation research has progressed from clean, continuously sounding targets toward intermittent goals, multiple competing sources, semantic generalization, and continuous motion [289], [290], [291], while active exploration explicitly treats movement as a sensing operation for improving separation, mapping, and acoustic modeling [94], [191], [293]. Autonomous driving remains less mature in its use of audition: although acoustic cues already improve traffic awareness, only recent models begin to connect audio directly with trajectory generation or broader world-action modeling [299], [301]. A central future direction is therefore to move from audio-assisted mobility toward genuinely omnimodal closed-loop policies that continuously decide where to move, which sensory evidence to acquire, how uncertain that evidence is, and how resulting actions should update subsequent perception and planning.

B. Manipulative Interaction

Audio-visual manipulative interaction concerns embodied settings in which an agent uses auditory and visual observations to physically alter the state of objects through grasping, pushing, insertion, pouring, wiping, tool use, or other manipulation actions. Unlike action recognition, which predicts what manipulation is being performed, manipulative interaction closes the perception–action loop by using multimodal feedback to determine how the robot should act and adapt during execution. Vision provides global information about object identity, pose, geometry, and motion, but often becomes unreliable once contact introduces occlusion or when relevant physical states are visually hidden. Interaction sounds, in contrast, can reveal high-frequency contact events, material properties, object states, and changes caused by force or friction. We organize existing work into *contact-rich manipulation*, where audio provides feedback about physical interaction itself, and *audio-visual skill execution*, where auditory and visual cues jointly support the completion and generalization of manipulation skills.

Methods. Existing audio-visual manipulation approaches can be broadly organized into two interaction paradigms: *contact-rich manipulation*, where auditory cues provide immediate feedback about physical contact and interaction dynamics, and *audio-visual skill execution*, where multimodal observations guide the completion and generalization of longer manipulation skills.

Contact-rich manipulation particularly benefits from audition because many decisive interaction events are difficult to observe visually. See, Hear, and Feel [302] demonstrates this complementarity by jointly fusing vision, contact audio, and tactile sensing for dense packing and pouring, where audio provides immediate cues for otherwise occluded interaction events. Auditory feedback can also serve as a direct supervision signal: That Sounds Right [303] collects large-scale contact sounds from dynamic manipulation and uses self-supervised acoustic representations to guide robot behaviors toward desired interaction profiles. Hearing Touch [304] further leverages large-scale audio-visual pretraining to initialize contact-microphone representations, allowing internet-scale audiovisual knowledge to improve low-data contact-rich manipulation. Rather than treating all modalities uniformly, Play to the Score [305] dynamically adjusts the importance of vision, audio, and touch according to the current stage of pouring and keyed insertion, reflecting the fact that acoustic evidence is often sparse but highly informative around critical contacts. Recent work increasingly seeks reusable physical representations: TacX [306] learns multisensory touch representations jointly from tactile images, audio, inertial measurements, and pressure over large-scale contact interactions, while hierarchical audio-visual-proprioceptive policies [307] explicitly condition manipulation actions on sparse contact-related acoustic cues. Emerging approaches even use generated sound as a proxy for contact force: Dreaming the Sound of Contact [308] combines generated video trajectories with audio-derived force profiles for closed-loop force-aware execution, suggesting that sound can encode not only whether contact occurs but also how strongly an interaction should be executed.

Beyond local contact regulation, audio-visual interaction can support complete manipulation skills by exposing task states that are difficult or impossible to infer from vision alone. Early systems combine visual motion and spoken or acoustic information for skill acquisition and robotic task execution [309], while audio-visual language-guided sorting [310] actively manipulates visually indistinguishable containers to generate diagnostic sounds before selecting the target object. Pouring is a particularly representative skill because liquid level and flow state are often only partially observable visually; visual-audio fusion has therefore been used to estimate pouring progress and directly control robot motion [311]. Larger multimodal resources such as RH20T [26], which records visual, force, audio, action, and language information across more than one hundred thousand contact-rich sequences, provide a foundation for learning broader manipulation skills. ManiWAV [27] further introduces an "ear-in-hand" interface for collecting synchronized in-the-wild audiovisual demonstrations and learns policies for tasks such as wiping, flipping, pouring, and Velcro manipulation, where sounds reveal contact modes, surface materials, or hidden object states. The Sound of Simulation [312] extends this paradigm to scalable sim-to-real learning by generating realistic audio conditioned on simulated video and training audiovisual policies for pouring that transfer

zero-shot to novel real-world containers and liquids. More recent acoustic-aware imitation learning frameworks such as S2A2 [313] additionally exploit both the spatial direction and spectral identity of sounds to select manipulation targets, indicating a shift from using audio merely as execution feedback toward treating it as an active source of task-relevant spatial and physical information.

Discussion. Audio-visual manipulation highlights a form of multimodal complementarity that is fundamentally different from passive perception: many useful acoustic signals are created *by the robot's own actions*. Contact, sliding, shaking, pouring, and tool use generate sounds that reveal material properties and latent object states, allowing manipulation policies to actively probe the world rather than merely observe it [310], [27]. Nevertheless, current systems remain predominantly task-specific and depend strongly on microphone placement, robot ego-noise, temporal calibration, and the acoustic properties of the surrounding environment. Interaction sounds are also sparse and highly action-dependent, requiring policies to combine slow, semantically rich visual observations with short but high-frequency auditory events [304], [305]. Recent progress in in-the-wild data collection, multisensory pretraining, and generative audiovisual simulation [26], [306], [312] provides promising routes toward more scalable manipulation learning. An important next step is to move from isolated audio-assisted skills toward policies that actively decide when to probe an object, which acoustic evidence is reliable, and how multimodal feedback should modify ongoing manipulation, while retaining the ability to transfer across objects, environments, materials, and previously unseen skills.

C. Communicative Interaction

Audio-visual communicative interaction concerns embodied settings in which humans and robots exchange verbal and non-verbal signals to establish shared understanding, coordinate behavior, and accomplish situated tasks. Unlike audio-visual question answering, where multimodal observations are mapped to a textual response without necessarily affecting the environment, communicative interaction forms a closed loop in which speech, prosody, gaze, gesture, facial or bodily behavior, and visual scene context influence subsequent robot responses and actions. Communication may therefore serve not only to convey explicit instructions but also to resolve ambiguous references, establish joint attention, negotiate task allocation, express uncertainty, or provide feedback during physical collaboration. We organize existing work into two task families: *situated dialogue*, where communication itself is the primary mechanism for grounding and coordinating interaction in a shared environment, and *human–robot collaboration*, where multimodal communication supports the execution of a shared physical task.

Methods. Existing approaches to communicative interaction increasingly move from predefined multimodal interfaces toward context-aware and bidirectional communication policies. Early systems explicitly fuse speech with gaze, gesture, or visual grounding to resolve human intent, whereas recent foundation-model-based approaches integrate scene understanding, dialogue history, and interaction state to determine not only how an utterance should be interpreted but also whether the robot should respond, ask for clarification, negotiate, or execute an action. According to the role of communication in the interaction loop, we group these approaches into situated dialogue and human–robot collaboration.

Situated dialogue requires linguistic communication to be continuously grounded in a shared physical and perceptual context. Early studies show that speech alone is insufficient for natural interaction: gaze and visual joint attention provide important cues for turn-taking, reference resolution, feedback, and task progression [314]. INGRESS [315] introduces interactive visual grounding in which a robot resolves natural referring expressions from language and visual observations and actively asks clarification questions when the intended object remains ambiguous. Subsequent work extends reference understanding with non-verbal communication; YouRefIt [316] demonstrates that gesture provides information complementary to language for embodied reference understanding, while audio-visual referring-expression grounding [317] directly integrates auditory and visual evidence into robotic manipulation instructions. Multisensory instruction understanding further incorporates auditory, visual, and tactile evidence before executing manipulation commands [318]. Recent systems increasingly replace hand-designed dialogue pipelines with foundation models. Vision-enabled dialogue on social robots [319] conditions conversational responses on live visual context, while explicit multimodal dialogue planning [320] enables an agent to generate clarification steps when visually grounded human commands are underspecified. NatSGLD [321] further provides paired speech, gesture, task logic, and demonstration trajectories for learning multimodal instruction following. More recent architectures selectively invoke visual grounding only when dialogue requires scene information [322], reflecting a shift from always-on multimodal fusion toward context-dependent perception and communication.

In human–robot collaboration, communication is embedded within a shared physical task and must support continuous coordination between partners. Co-speech interaction systems combine spoken instructions with visual gestures to assign objects and robot actions in collaborative workspaces [323], while multimodal robot interfaces increasingly use speech together with gaze or body cues to reduce ambiguity and improve task execution. In industrial settings, gesture- and voice-controlled robot trajectories have been shown to provide more intuitive intervention during collaborative operations [324], and bidirectional verbal systems integrate speech recognition, visual object detection, text-to-speech, and robot control to support task coordination and quality inspection [325]. Recent work increasingly emphasizes communication as an adaptive two-way process rather than a sequence of human commands. BRIDGE [326] allows users to modify robot position, velocity, and force through natural-language dialogue while the robot verbally confirms changes or requests clarification, directly coupling communication with physical assistance. Similarly, speech-driven collaborative inspection systems combine visual perception, local language models, and robot control to translate spoken requests into task-specific motion plans while providing verbal feedback [327]. Going beyond re-

active dialogue, MICoBot [328] introduces mixed-initiative communication for long-horizon collaborative manipulation, allowing both human and robot to propose, accept, or reject task allocations according to robot capability and estimated human willingness. These developments mark a progression from multimodal command interfaces toward collaborative agents that use dialogue to dynamically negotiate how a shared physical task should be accomplished.

Discussion. Communicative interaction exposes requirements that are largely absent from offline multimodal reasoning: successful communication depends not only on semantic correctness but also on timing, common ground, perspective, uncertainty, and the evolving physical state of a shared task. Speech may be ambiguous or corrupted by environmental noise, while gaze and gesture can be incomplete, conflicting, or difficult to perceive; consequently, robust agents must determine which communicative cues are reliable and when clarification is preferable to immediate action [315], [320]. Recent systems increasingly address this challenge through visually grounded dialogue, multimodal instruction datasets, and bidirectional feedback [319], [321], [326]. Human–robot collaboration further requires communication to adapt to changing capabilities and responsibilities during execution, motivating mixed-initiative systems in which both partners can influence task allocation and behavior [328]. An important future direction is therefore to move beyond command-following interfaces toward continuously grounded communicative policies that integrate speech, prosody, gaze, gesture, and physical feedback, explicitly represent uncertainty and common ground, and decide when to speak, listen, clarify, or act as part of a unified closed-loop interaction process.

D. Generalist Policies

Definition. Generalist policies aim to replace task-specific controllers with a shared multimodal policy that can execute diverse behaviors across tasks, objects, environments, sensory configurations, or robot embodiments. Unlike mobility, manipulation, and communicative interaction, which categorize agents according to the form of interaction they perform, generalist policies constitute a cross-cutting capability that concerns how broadly a single policy can reuse its perceptual and control knowledge. A generalist policy therefore maps heterogeneous observations and task specifications to physical actions while retaining the ability to transfer or adapt beyond the conditions on which it was originally trained. In the context of embodied audio-visual intelligence, this requires moving beyond conventional vision–language–action models toward policies that can additionally exploit speech, environmental sound, and interaction-generated acoustic feedback. We organize this emerging direction into *audio-visual-language policies*, which incorporate audition directly into language-conditioned control, and *cross-task policy generalization*, which seeks reusable policies across increasingly diverse tasks, environments, embodiments, and sensing configurations.

Methods. The development of generalist embodied policies follows two closely related trends. One expands the sensory interface of vision–language–action models so that sound becomes a first-class observation for closed-loop control; the other scales policy pretraining across heterogeneous robot data to improve transfer across tasks and embodiments. While most early generalist robot policies are primarily vision–language based, they provide the policy architectures and large-scale pretraining substrates on which recent audio-enabled models are built.

Audio-Visual-Language Policies. Early generalist robot models establish the basic vision–language–action paradigm by transferring semantic knowledge from pretrained language or vision-language models to robotic control [329], [330], [331]. However, their sensory interface remains dominated by vision and proprioception, leaving physical states that are easier to hear than to see largely unobserved. Beyond Sight, or FuSe [332], directly addresses this limitation by fine-tuning generalist policies with heterogeneous sensing, including vision, touch, and audio, while using language as a common grounding space; the same formulation is demonstrated on both diffusion-based generalist policies and VLA models. More recent systems move from auxiliary acoustic prompting toward native auditory control. Audio-VLA [333] integrates contact audio into a VLA policy so that transient interaction events and manipulation dynamics can influence action generation, whereas SVA [334] replaces purely textual instructions with speech and jointly processes speech, vision, and proprioception for real-time robotic control. HEAR [335] further formalizes the Vision-Sound-Language-Action paradigm by continuously conditioning control on streaming sound and introducing acoustic memory to preserve brief events that may occur between action chunks. At a broader physical-AI scale, Cosmos 3 [301] jointly represents language, images, video, audio, and actions within a unified omnimodal world model, suggesting a future in which sensing, prediction, simulation, and policy execution can share the same multimodal representation rather than relying on separate perception and control modules.

Cross-Task Policy Generalization. A complementary direction seeks to make a single policy reusable across increasingly heterogeneous robotic experiences. Open X-Embodiment [336] demonstrates that pooling trajectories from many robot platforms can produce positive cross-embodiment transfer, while Octo [337] develops an openly available generalist policy that supports different robots, observations, task specifications, and action spaces through a shared Transformer policy. OpenVLA [330] and π_0 [331] further scale this paradigm through pretrained vision-language representations and large, heterogeneous robot datasets, enabling zero-shot execution and efficient adaptation to new manipulation skills. Recent work explores alternative sources of generalization: Video Prediction Policy [338] uses predictive representations from video foundation models to capture physical dynamics across tasks, while RoboArena [339] highlights the need to evaluate generalist policies across distributed real-world environments rather than a single standardized laboratory setup. Importantly for audio-visual intelligence, FuSe [332] shows that generalization should extend not only across tasks and embodiments but also across *sensor modalities*: a pretrained policy can acquire scarce auditory or tactile capabilities

without rebuilding the complete policy from scratch. Recent analyses further show that large heterogeneous datasets do not automatically guarantee robust generalization, as dataset fragmentation and shortcut learning can still cause policies to rely on spurious task-specific correlations [340]. Generality should therefore be understood along multiple axes—task, object, environment, embodiment, and sensor modality—rather than merely by the number of behaviors demonstrated by a single model.

Discussion. Generalist policies represent a transition from assembling separate controllers for navigation, manipulation, and communication toward reusable embodied models that share perception and action knowledge across tasks. Nevertheless, today's strongest generalist robot policies remain predominantly vision–language based, and genuine audio-visual-language-action learning is still comparatively immature. Recent systems demonstrate that audition can contribute information unavailable to vision, ranging from spoken instructions to short contact events and latent physical states [332], [333], [334], [335], yet several challenges remain: audio is temporally dense and asynchronous with visual policy updates, interaction sounds depend strongly on the robot's own actions, and large-scale robot datasets rarely contain synchronized high-quality acoustic observations. Moreover, broad task coverage does not necessarily imply robust generalization across new environments, embodiments, or sensing conditions [339], [340]. An important future direction is therefore to develop genuinely omnimodal generalist policies that learn from heterogeneous robot and Internet-scale multimodal data, dynamically select useful sensory evidence, maintain audio-visual context over long-horizon execution, and transfer shared world and action knowledge across manipulation, mobility, and communicative interaction.

VI. Further Discussion and Future Directions

The Perception–Reasoning–Interaction taxonomy reveals a central gap in audio-visual embodied intelligence: improvements in recognizing multimodal content or answering questions do not directly establish improvements in closed-loop behavior. An agent must preserve sensory evidence as it moves from state estimation to inference and action, and use subsequent observations to revise its beliefs. Building on the preceding task-level discussions, we identify six interconnected directions toward this objective. Across these directions, the key question is under what conditions auditory information changes a decision and whether that change improves embodied performance.

A. Hallucination and Evidence-Grounded Decision-Making

Audio-visual hallucination arises when a model asserts events, entities, or cross-modal relationships that are unsupported by the available observations. A visible object may induce the prediction of a characteristic sound even when it is silent, while an audible event may be incorrectly attributed to a plausible visible source. AVHBench demonstrates that interactions between modalities can themselves introduce hallucinations [341]. In embodied settings, such errors can propagate beyond language: an incorrect source association may redirect navigation, and an inferred contact event may prematurely terminate manipulation. Nevertheless, cross-modal disagreement is not necessarily erroneous. Off-screen sources, occlusion, and delayed acoustic responses can produce valid observations that do not coincide across modalities. Reliable systems must therefore distinguish unsupported assertions from uncertain but physically plausible hypotheses, rather than enforce unconditional agreement between audio and vision.

A promising direction is to preserve the provenance and uncertainty of evidence throughout the perception–reasoning–action loop. World-state estimates and consequential reasoning steps should remain associated with supporting acoustic intervals, visual entities, observation times, and coordinate frames. Process supervision, as explored in video-SALMONN-o1 [245], provides a starting point, but a fluent intermediate explanation alone does not establish sensory faithfulness. Future training should test whether controlled changes to relevant evidence induce appropriate changes in conclusions, while irrelevant changes leave them stable. At execution time, uncertainty should influence whether the agent proceeds, gathers another observation, or requests clarification, extending the principle of interactive grounding [315]. Evaluation should connect unsupported beliefs to their downstream consequences and measure calibrated confidence, unnecessary interventions, and recovery after contradictory feedback.

B. Action-Conditioned World Models and Active Sensing

The relationship between sound and vision changes when an agent can intervene in the scene. A tap, a change of viewpoint, or the opening of a door alters both the physical state and the evidence available for subsequent inference. Physical audio-visual commonsense benchmarks such as PACS [24] assess reasoning about latent properties, whereas Move2Hear and ActiveRIR demonstrate how movement can improve auditory perception and acoustic modeling [94], [293]. These capabilities motivate a shared, action-conditioned world representation that links scene geometry, object properties, sound generation, and propagation. Omnimodal models such as Cosmos 3 provide an emerging framework for jointly modeling language, vision, audio, and actions [301]; however, modality coverage alone does not establish that a model has learned the physical mechanisms needed for reliable acoustic prediction and control.

Future world models should predict how candidate actions change both the environment and its observable audio-visual consequences. An important distinction is between a change in the source, such as altered contact force, and a change in transmission, such as a different source–microphone distance. Conflating these factors can produce incorrect physical inferences even when generated audio sounds plausible. Learning should therefore combine controlled interventions with geometric and acoustic constraints, and assess predictions under changes in material, action, viewpoint, and room acoustics. Such models could support active sensing by comparing the expected decision benefit of moving, waiting, probing an object, or asking a question against the associated time, energy,

and interaction costs. The objective is task-relevant uncertainty reduction: acquiring a clearer sound is useful when it improves the next decision, rather than merely increasing reconstruction fidelity.

C. Streaming Fusion, Memory, and Real-Time Control

Embodied audio-visual intelligence requires causal processing of continuous observations under finite computation. Brief impacts or changes in friction may occur between visual updates, while spoken instructions and environmental events unfold over longer intervals. Qwen2.5-Omni explores temporally aligned multimodal streaming [254], and video-SALMONN S investigates memory for extended audio-visual streams [267]. However, streaming understanding and timely motor feedback impose different requirements. A model may process long recordings successfully while still overlooking a transient event or responding too late to modify an ongoing action. Real-time capability must consequently be assessed from the arrival of decision-relevant evidence to the resulting physical response, including buffering, encoding, inference, communication, and actuation delays.

A promising architecture combines rapid event-sensitive updates with slower semantic reasoning and longer-term episodic memory. Rather than force every modality into a common sampling rate, the system should retain explicit timestamps and selectively increase processing around informative events. Memory should preserve source identity, agent pose, action context, and uncertainty alongside compact semantic summaries; retaining access to selected raw intervals can help recover acoustic details lost during compression. Egocentric memory systems such as EgoLife [271] offer useful foundations, but action-oriented memory must additionally distinguish previously observed states from states that remain valid after intervention. Policies should also be able to interrupt or revise action sequences when new sensory evidence invalidates their assumptions. Suitable evaluation should measure missed-event rates, evidence-to-action latency, memory cost, and recovery performance under strictly causal observation, rather than infer deployability from average inference speed alone.

D. Scalable Data and Physically Consistent Sim-to-Real Transfer

Audio-visual policy learning requires data that connect observations to the actions that produced them and the outcomes that followed. RH20T and ManiWAV provide complementary foundations through multimodal robot recordings and in-the-wild audio-visual demonstrations [26], [27]. Scaling this supervision remains difficult because acoustic observations depend on microphone placement, sensor response, room propagation, robot motion, and object interaction. These factors can become entangled with task identity, allowing a policy to exploit recording conditions instead of transferable physical cues. Future datasets should therefore document timing and calibration, preserve spatial audio where available, and include unsuccessful attempts, corrective actions, quiet interactions, and irrelevant sounds. Matched recordings across rooms, objects, and sensor configurations would help separate task information from acquisition-specific correlations.

Simulation offers a complementary route to coverage, provided that synthetic sound preserves the physical distinctions needed for action. The Sound of Simulation demonstrates generative audio augmentation for sim-to-real audiovisual pouring policies [312]. Extending this approach requires more than perceptual realism: generated audio should respond consistently to changes in contact, velocity, material, and source–receiver geometry. A useful direction is to combine physical simulation, learned sound generation, and acoustic-field estimation [200], while varying source dynamics and propagation conditions separately. Small amounts of real interaction data could then calibrate the remaining mismatch. Transfer should be tested on held-out combinations of object properties, room acoustics, microphone configurations, and robot embodiments. Crucially, policy evaluation must establish whether synthetic acoustic cues improve real-world decisions or instead introduce shortcuts that disappear after deployment.

E. Generalist Audio-Visual-Language-Action Policies

Generalist policies provide a potential interface through which perceptual and reasoning capabilities can be reused across mobility, manipulation, and communication. For audio-visual intelligence, this requires distinguishing the roles of speech, environmental sound, and action-generated feedback. A spoken instruction specifies a goal; an off-screen sound supplies evidence about the surroundings; a contact transient can reveal the outcome of an executed action. Treating these signals as interchangeable text descriptions risks losing timing, spatial structure, and physical information. FuSe demonstrates that heterogeneous sensors, including audio, can be incorporated into pretrained generalist policies through language grounding [332]. This offers a route for transferring semantic knowledge, while leaving open how to preserve fine-grained acoustic evidence throughout action generation.

Future policies should combine shared task and world representations with sensor- and embodiment-aware interfaces. Internet audio-visual data can support semantic pretraining, but robot interaction data are needed to connect sound to controllable physical consequences. Training should explicitly discourage visual or linguistic shortcuts and reward the use of sound when it resolves an otherwise ambiguous decision. At the same time, a policy should remain effective when audio is unavailable, uninformative, or corrupted; accepting an additional modality is insufficient if its influence cannot be calibrated. Generalization should be evaluated separately across tasks, objects, environments, embodiments, and sensor configurations, including combinations absent from training. The longer-term objective is a shared policy that can navigate toward an acoustic event, inspect its source, manipulate an object, and communicate uncertainty while maintaining a consistent account of what has been observed and what its actions have changed.

F. From Benchmark Accuracy to Closed-Loop Evaluation

A convincing evaluation framework must connect the three capability dimensions without conflating their outputs. Per-

ceptual localization accuracy, reasoning correctness, and task completion measure different properties, and success at one level may conceal failure at another. Bias analyses in audio-visual question answering illustrate why high scores need not imply appropriate evidence use [225], [226]. Future benchmarks should combine component diagnostics with end-to-end episodes that require the agent to acquire evidence, revise a belief, act, and verify the outcome. Audio-only, visual-only, transcript-only, and joint-input baselines should be accompanied by targeted interventions on source identity, timing, and spatial correspondence. Test-time removal measures sensitivity to missing input, whereas separately trained unimodal baselines help assess the benefit of adding a modality; neither comparison alone establishes causal understanding. Physically realizable interventions should complement artificial mismatches, whose effects may otherwise reflect distribution shift.

Evaluation should also separate improvements due to acoustic information from those due to additional model capacity, tool calls, or inference time. Matched resource budgets and repeated trials should report task success, action efficiency, failure severity, recovery, and latency alongside evidence-grounding metrics. Distributed real-world evaluation, exemplified by RoboArena [339], provides a useful model for testing policies beyond a single laboratory, although an audio-visual extension would need explicit control and documentation of acoustic conditions. Training and test splits should additionally separate recording sessions, environments, and embodiments to limit leakage through background acoustics or device signatures. Such protocols would establish whether an agent uses sound selectively and reliably, and whether its perceptual and reasoning improvements persist when actions change the observations it receives.

VII. Conclusion

References

[1] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat *et al.*, "Gpt-4 technical report," *arXiv preprint arXiv:2303.08774*, 2023.

[2] G. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican *et al.*, "Gemini: a family of highly capable multimodal models," *arXiv preprint arXiv:2312.11805*, 2023.

[3] Y. Chu, J. Xu, X. Zhou, Q. Yang, S. Zhang, Z. Yan, C. Zhou, and J. Zhou, "Qwen-audio: Advancing universal audio understanding via unified large-scale audio-language models," *arXiv preprint arXiv:2311.07919*, 2023.

[4] R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra, "Imagebind one embedding space to bind them all," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023, pp. 15 180–15 190.

[5] H. Zhu, M.-D. Luo, R. Wang, A.-H. Zheng, and R. He, "Deep audio-visual learning: A survey," *International Journal of Automation and Computing*, vol. 18, no. 3, pp. 351–376, 2021. [Online]. Available: https://link.springer.com/article/10.1007/s11633-021-1293-0

[6] Y. Wei, D. Hu, Y. Tian, and X. Li, "Learning in audio-visual context: A review, analysis, and new perspective," *arXiv preprint arXiv:2208.09579*, 2022. [Online]. Available: https://arxiv.org/abs/2208.09579

[7] A. Shahabaz and S. Sarkar, "Increasing importance of joint analysis of audio and video in computer vision: A survey," *IEEE Access*, vol. 12, pp. 59 399–59 430, 2024.

[8] M. AlSuwat, S. Al-Shareef, and M. AlGhamdi, "Audio-visual self-supervised representation learning: A survey," *Neurocomputing*, vol. 634, p. 129750, 2025.

[9] J. Zhao, Y. Xu, X. Qian, D. Berghi, P. Wu, M. Cui, J. Sun, P. J. B. Jackson, and W. Wang, "Audio-visual speaker tracking: Progress, challenges, and future directions," *arXiv preprint arXiv:2310.14778*, 2023. [Online]. Available: https://arxiv.org/abs/2310.14778

[10] Y. Wu, P. Zhang, M. Gu, J. Zheng, and X. Bai, "Embodied navigation with multi-modal information: A survey from tasks to methodology," *Information Fusion*, vol. 112, p. 102532, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253524003105

[11] J. Li and Y. Tian, "From waveforms to pixels: A survey on audio-visual segmentation," *arXiv preprint arXiv:2508.03724*, 2025. [Online]. Available: https://arxiv.org/abs/2508.03724

[12] W. You, X. Diao, W. Huang, C. Zhang, K. Kong, W. Wu, C. Ma, Z. Ouyang, T. Wu, M. Cheng, S. Vosoughi, and J. Gui, "Music audio-visual question answering requires specialized multimodal designs," in *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, 2026, pp. 1392–1426. [Online]. Available: https://aclanthology.org/2026.findings-acl.69/

[13] Y. Qin, K. Liu, S. Wu, K. Wang, S. Deng, Y. Tian, J. Xiao, Y. Xing, Y. Ma, B. Li, R. Zimmermann, L. Cui, F. Wei, J. Luo, and H. Fei, "Audio-visual intelligence in large foundation models: A comprehensive survey," *arXiv preprint arXiv:2605.04045*, 2026. [Online]. Available: https://arxiv.org/abs/2605.04045

[14] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audio set: An ontology and human-labeled dataset for audio events," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 776–780.

[15] H. Zhao, C. Gan, A. Rouditchenko, C. Vondrick, J. McDermott, and A. Torralba, "The sound of pixels," in *Proceedings of the European Conference on Computer Vision*, 2018, pp. 570–586.

[16] Y. Tian, J. Shi, B. Li, Z. Duan, and C. Xu, "Audio-visual event localization in unconstrained videos," in *Computer Vision – ECCV 2018*. Springer, 2018, pp. 252–268.

[17] H. Chen, W. Xie, A. Vedaldi, and A. Zisserman, "VGGSound: A large-scale audio-visual dataset," in *2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 721–725.

[18] Y. Tian, D. Li, and C. Xu, "Unified multisensory perception: Weakly-supervised audio-visual video parsing," in *Computer Vision – ECCV 2020*. Springer, 2020, pp. 436–454.

[19] D. Damen, H. Doughty, G. M. Farinella, A. Furnari, J. Ma, E. Kazakos, D. Moltisanti, J. Munro, T. Perrett, W. Price, and M. Wray, "Rescaling egocentric vision: Collection, pipeline and challenges for EPIC-KITCHENS-100," *International Journal of Computer Vision*, vol. 130, pp. 33–55, 2022.

[20] H. Chen, W. Xie, T. Afouras, A. Nagrani, A. Vedaldi, and A. Zisserman, "Localizing visual sounds the hard way," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2021, pp. 16 862–16 871.

[21] J. Zhou, J. Wang, J. Zhang, W. Sun, J. Zhang, S. Birchfield, D. Guo, L. Kong, M. Wang, and Y. Zhong, "Audio–visual segmentation," in *European Conference on Computer Vision*. Springer, 2022, pp. 386–403.

[22] H. Yun, Y. Yu, W. Yang, K. Lee, and G. Kim, "Pano-AVQA: Grounded audio-visual question answering on 360° videos," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 2031–2041. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/html/Yun_Pano-AVQA_Grounded_Audio-Visual_Question_Answering_on_360deg_Videos_ICCV_2021_paper.html

[23] G. Li, Y. Wei, Y. Tian, C. Xu, J.-R. Wen, and D. Hu, "Learning to answer questions in dynamic audio-visual scenarios," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 19 108–19 118. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Li_Learning_To_Answer_Questions_in_Dynamic_Audio-Visual_Scenarios_CVPR_2022_paper.html

[24] S. Yu, P. Wu, P. P. Liang, R. Salakhutdinov, and L.-P. Morency, "PACS: A dataset for physical audiovisual CommonSense reasoning," in *Computer Vision–ECCV 2022*. Springer, 2022, pp. 292–309. [Online]. Available: https://doi.org/10.1007/978-3-031-19836-6_17

[25] C. Chen, U. Jain, C. Schissler, S. V. Amengual Gari, Z. Al-Halah, V. K. Ithapu, P. Robinson, and K. Grauman, "SoundSpaces: Audio-visual navigation in 3d environments," in *Computer Vision – ECCV 2020*. Springer, 2020, pp. 17–36.

[26] H.-S. Fang, H. Fang, Z. Tang, J. Liu, C. Wang, J. Wang, H. Zhu, and C. Lu, "RH20T: A comprehensive robotic dataset for learning diverse skills in one-shot," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 653–660.

[27] Z. Liu, C. Chi, E. Cousineau, N. Kuppuswamy, B. Burchfiel, and S. Song, "ManiWAV: Learning robot manipulation from in-the-wild audio-visual data," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 947–962.

[28] F. Xiao, Y. J. Lee, K. Grauman, J. Malik, and C. Feichtenhofer, "Audiovisual slowfast networks for video recognition," *arXiv preprint arXiv:2001.08740*, 2020.

[29] H. Akbari, L. Yuan, R. Qian, W.-H. Chuang, S.-F. Chang, Y. Cui, and B. Gong, "Vatt: Transformers for multimodal self-supervised learning from raw video, audio and text," *Advances in neural information processing systems*, vol. 34, pp. 24 206–24 221, 2021.

[30] Y. Aytar, C. Vondrick, and A. Torralba, "Soundnet: Learning sound representations from unlabeled video," *Advances in neural information processing systems*, vol. 29, 2016.

[31] Y. Gong, A. Rouditchenko, A. H. Liu, D. Harwath, L. Karlinsky, H. Kuehne, and J. R. Glass, "Contrastive audio-visual masked autoencoder," in *The Eleventh International Conference on Learning Representations*, 2023.

[32] K. Yang, B. Russell, and J. Salamon, "Telling left from right: Learning spatial correspondence of sight and sound," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 9932–9941.

[33] P. Morgado, Y. Li, and N. Vasconcelos, "Learning representations from audio-visual spatial alignment," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 4733–4744.

[34] A. Nagrani, S. Yang, A. Arnab, A. Jansen, C. Schmid, and C. Sun, "Attention bottlenecks for multimodal fusion," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 14 200–14 213.

[35] R. Arandjelovic and A. Zisserman, "Look, listen and learn," in *2017 IEEE International Conference on Computer Vision (ICCV)*. IEEE, 2017, pp. 609–617.

[36] J. S. Chung and A. Zisserman, "Out of time: Automated lip sync in the wild," in *Workshop on Multi-view Lip-reading, Asian Conference on Computer Vision (ACCV)*, 2016.

[37] B. Korbar, D. Tran, and L. Torresani, "Cooperative learning of audio and video models from self-supervised synchronization," *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[38] A. Owens, P. Isola, J. McDermott, A. Torralba, E. H. Adelson, and W. T. Freeman, "Visually indicated sounds," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2405–2413.

[39] A. Owens, J. Wu, J. H. McDermott, W. T. Freeman, and A. Torralba, "Ambient sound provides supervision for visual learning," in *European conference on computer vision*. Springer, 2016, pp. 801–816.

[40] R. Arandjelovic and A. Zisserman, "Objects that sound," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 435–451.

[41] A. L. Cramer, H.-H. Wu, J. Salamon, and J. P. Bello, "Look, listen, and learn more: Design choices for deep audio embeddings," in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 3852–3856.

[42] D. Hu, F. Nie, and X. Li, "Deep multimodal clustering for unsupervised audiovisual learning," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2019, pp. 9240–9249.

[43] H. Alwassel, D. Mahajan, B. Korbar, L. Torresani, B. Ghanem, and D. Tran, "Self-supervised learning by cross-modal audio-video clustering," *Advances in neural information processing systems*, vol. 33, pp. 9758–9770, 2020.

[44] Y. Asano, M. Patrick, C. Rupprecht, and A. Vedaldi, "Labelling unlabelled videos from scratch with multi-modal self-supervision," *Advances in Neural Information Processing Systems*, vol. 33, pp. 4660–4671, 2020.

[45] P. Morgado, N. Vasconcelos, and I. Misra, "Audio-visual instance discrimination with cross-modal agreement," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2021, pp. 12 470–12 481.

[46] P. Morgado, I. Misra, and N. Vasconcelos, "Robust audio-visual instance discrimination," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2021, pp. 12 929–12 940.

[47] J.-B. Alayrac, A. Recasens, R. Schneider, R. Arandjelović, J. Ramapuram, J. De Fauw, L. Smaira, S. Dieleman, and A. Zisserman, "Self-supervised multimodal versatile networks," *Advances in neural information processing systems*, vol. 33, pp. 25–37, 2020.

[48] Y. Chen, Y. Xian, A. Koepke, Y. Shan, and Z. Akata, "Distilling audio-visual knowledge by compositional contrastive learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 7016–7025.

[49] M.-I. Georgescu, E. Fonseca, R. T. Ionescu, M. Lucic, C. Schmid, and A. Arnab, "Audiovisual masked autoencoders," in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2023, pp. 16 098–16 108.

[50] P.-Y. Huang, V. Sharma, H. Xu, C. Ryali, Y. Li, S.-W. Li, G. Ghosh, J. Malik, C. Feichtenhofer *et al.*, "Mavil: Masked audio-video learners," *Advances in neural information processing systems*, vol. 36, pp. 20 371–20 393, 2023.

[51] A. Guzhov, F. Raue, J. Hees, and A. Dengel, "Audioclip: Extending clip to image, text and audio," in *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2022, pp. 976–980.

[52] B. Zhu, B. Lin, M. Ning, Y. Yan, J. Cui, W. HongFa, Y. Pang, W. Jiang, J. Zhang, Z. Li *et al.*, "Languagebind: Extending video-language pretraining to n-modality by language-based semantic alignment," in *International Conference on Learning Representations*, vol. 2024, 2024, pp. 9588–9608.

[53] A. Vyas, H.-J. Chang, C.-F. Yang, P.-Y. Huang, L. Gao, J. Richter, S. Chen, M. Le, P. Dollár, C. Feichtenhofer *et al.*, "Pushing the frontier of audiovisual perception with large-scale multimodal correspondence learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 30 172–30 182.

[54] S. Lee, J. Chung, Y. Yu, G. Kim, T. Breuel, G. Chechik, and Y. Song, "Acav100m: Automatic curation of large-scale datasets for audio-visual video representation learning," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2021, pp. 10 254–10 264.

[55] P. Sarkar and A. Etemad, "Self-supervised audio-visual representation learning with relaxed cross-modal synchronicity," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 37, no. 8, 2023, pp. 9723–9732.

[56] A. Senocak, H. Ryu, J. Kim, T.-H. Oh, H. Pfister, and J. S. Chung, "Sound source localization is all about cross-modal alignment," in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2023, pp. 7743–7753.

[57] M. Slaney and M. Covell, "Facesync: A linear operator for measuring synchronization of video facial images and audio tracks," *Advances in Neural Information Processing Systems*, vol. 13, 2000.

[58] A. Owens and A. A. Efros, "Audio-visual scene analysis with self-supervised multisensory features," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 631–648.

[59] N. Khosravan, S. Ardeshir, and R. Puri, "On attention modules for audio-visual synchronization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019, pp. 25–28.

[60] T. Kikuchi and Y. Ozasa, "Watch, listen once, and sync: Audio-visual synchronization with multi-modal regression cnn," in *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2018, pp. 3036–3040.

[61] S.-W. Chung, J. S. Chung, and H.-G. Kang, "Perfect match: Improved cross-modal embeddings for audio-visual synchronisation," in *2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 3965–3969.

[62] Y. J. Kim, H. S. Heo, S.-W. Chung, and B.-J. Lee, "End-to-end lip synchronisation based on pattern classification," in *2021 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2021, pp. 598–605.

[63] V. S. Kadandale, J. F. Montesinos, and G. Haro, "Vocalist: An audio-visual synchronisation model for lips and voices," in *Interspeech 2022*, 2022, pp. 3128–3132.

[64] X. Chen, H. Wu, C.-C. Wang, H.-Y. Lee, and J.-S. R. Jang, "Multimodal transformer distillation for audio-visual synchronization," in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 7755–7759.

[65] J. Lee, S.-W. Chung, S. Kim, H.-G. Kang, and K. Sohn, "Looking into your speech: Learning cross-modal affinity for audio-visual speech separation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 1336–1345.

[66] K. R. Prajwal, R. Mukhopadhyay, V. P. Namboodiri, and C. V. Jawahar, "A lip sync expert is all you need for speech to lip generation in the wild," in *Proceedings of the 28th ACM International Conference on Multimedia*, 2020, pp. 484–492.

[67] Z. Niu and B. Mak, "On the audio-visual synchronization for lip-to-speech synthesis," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 7843–7852.

[68] H. Chen, W. Xie, T. Afouras, A. Nagrani, A. Vedaldi, and A. Zisserman, "Audio-visual synchronisation in the wild," in *British Machine Vision Conference (BMVC)*, 2021.

[69] V. Iashin, W. Xie, E. Rahtu, and A. Zisserman, "Sparse in space and time: Audio-visual synchronisation with trainable selectors," in *British Machine Vision Conference (BMVC)*, 2022.

[70] Z. J. Liu, D. Chen, Y. Feng, Y. Ling, S. Luo, S. Dong, and B. Ferry, "Detect audio-video temporal synchronization errors in advertisements (ads)," in *2022 26th International Conference on Pattern Recognition (ICPR)*. IEEE, 2022, pp. 16–22.

[71] V. Iashin, W. Xie, E. Rahtu, and A. Zisserman, "Synchformer: Efficient synchronization from sparse cues," in *2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024.

[72] S. Jenni, A. Black, and J. Collomosse, "Audio-visual contrastive learning with temporal self-supervision," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 7, 2023, pp. 7996–8004.

[73] J. Wang, J. Jiao, Y. Song, S. James, Z. Tong, C. Ge, P. Abbeel, and Y.-H. Liu, "Speed co-augmentation for unsupervised audio-visual pre-training," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop on Sight and Sound (CVPRW)*, 2023.

[74] L. Goncalves, P. Mathur, C. Lavania, M. Cekic, M. Federico, and K. J. Han, "Perceptual evaluation of audio-visual synchrony grounded in viewers' opinion scores," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024, pp. 288–305.

[75] J. G. Voas, W.-C. Tseng, L. Berry, X. Hu, P. Peng, J. Stuedemann, and D. Harwath, "Temporally streaming audio-visual synchronization for real-world videos," in *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 5009–5017.

[76] A. Anshul, S. Gopal, D. Rajan, and E. S. Chng, "Intra-modal and cross-modal synchronization for audio-visual deepfake detection and temporal localization," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 13 826–13 836.

[77] H. K. Cheng, M. Ishii, A. Hayakawa, T. Shibuya, A. Schwing, and Y. Mitsufuji, "Mmaudio: Taming multimodal joint training for high-quality video-to-audio synthesis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 28 901–28 911.

[78] T. Sato, Y. Sugano, and Y. Sato, "Self-supervised learning for audio-visual relationships of videos with stereo sounds," *IEEE Access*, vol. 10, pp. 94 273–94 284, 2022.

[79] P. Morgado, N. Vasconcelos, T. Langlois, and O. Wang, "Self-supervised generation of spatial audio for 360° video," in *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[80] R. Gao and K. Grauman, "2.5d visual sound," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 324–333.

[81] Y.-B. Lin and Y.-C. F. Wang, "Exploiting audio-visual consistency with partial supervision for spatial audio generation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 3, 2021, pp. 2056–2063.

[82] X. Xu, H. Zhou, Z. Liu, B. Dai, X. Wang, and D. Lin, "Visually informed binaural audio generation without binaural audios," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 15 485–15 494.

[83] K. K. Rachavarapu, Aakanksha, V. Sundaresha, and A. N. Rajagopalan, "Localize to binauralize: Audio spatialization from visual sound source localization," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 1930–1939.

[84] R. Garg, R. Gao, and K. Grauman, "Geometry-aware multi-task learning for binaural audio generation from video," in *British Machine Vision Conference*, 2021.

[85] Z. Li, B. Zhao, and Y. Yuan, "Cyclic learning for binaural audio generation and localization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 26 669–26 678.

[86] Y. Chen, K. Shimada, C. Simon, Y. Ikemiya, T. Shibuya, and Y. Mitsufuji, "CCStereo: Audio-visual contextual and contrastive learning for binaural audio generation," in *Proceedings of the ACM International Conference on Multimedia*, 2025, pp. 7510–7518.

[87] S. Majumder, Z. Al-Halah, and K. Grauman, "Learning spatial features from audio-visual correspondence in egocentric videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 058–27 068.

[88] A. Min, Z. Chen, H. Zhao, and A. Owens, "Supervising sound localization by in-the-wild egomotion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 23 936–23 946.

[89] W. Guo, C. Pan, Z. Zhu, X. Hu, Y. Zhang, L. Tang, R. Yang, H. Wang, Z. Zhang, Y. Wang *et al.*, "Mrsaudio: A large-scale multimodal recorded spatial audio dataset with refined annotations," vol. 38, 2026.

[90] C. Gan, Y. Zhang, J. Wu, B. Gong, and J. B. Tenenbaum, "Look, listen, and act: Towards audio-visual embodied navigation," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 9701–9707.

[91] C. Chen, S. Majumder, A.-H. Ziad, R. Gao, S. Kumar Ramakrishnan, and K. Grauman, "Learning to set waypoints for audio-visual navigation," in *ICLR*, 2021.

[92] C. Chen, Z. Al-Halah, and K. Grauman, "Semantic audio-visual navigation," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2021, pp. 15 511–15 520.

[93] H. Mittal, P. Morgado, U. Jain, and A. Gupta, "Learning state-aware visual representations from audible interactions," *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[94] S. Majumder, Z. Al-Halah, and K. Grauman, "Move2Hear: Active audio-visual source separation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 275–285.

[95] A. Senocak, T.-H. Oh, J. Kim, M.-H. Yang, and I. S. Kweon, "Learning to localize sound source in visual scenes," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, 2018, pp. 4358–4366.

[96] S. Mo and P. Morgado, "Localizing visual sounds the easy way," in *European Conference on Computer Vision*. Springer, 2022, pp. 218–234.

[97] H. Xuan, Z. Wu, J. Yang, Y. Yan, and X. Alameda-Pineda, "A proposal-based paradigm for self-supervised sound source localization in videos," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2022, pp. 1019–1028.

[98] T. Afouras, Y. M. Asano, F. Fagan, A. Vedaldi, and F. Metze, "Self-supervised object detection from audio-visual correspondence," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2022, pp. 10 565–10 576.

[99] S. Mo and Y. Tian, "Audio-visual grouping network for sound localization from mixtures," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023, pp. 10 565–10 574.

[100] T. Mahmud, Y. Tian, and D. Marculescu, "T-vsl: Text-guided visual sound source localization in mixtures," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 26 732–26 741.

[101] S. J. Um, D. Kim, S. Lee, and J. U. Kim, "Object-aware sound source localization via audio-visual scene understanding," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 8342–8351.

[102] I. Kim, Y. Song, J. Park, W. H. Kim, and S. Kwak, "Improving sound source localization with joint slot attention on image and audio," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 3121–3130.

[103] H. Ryu, S. Kim, J. S. Chung, and A. Senocak, "Seeing speech and sound: Distinguishing and locating audio sources in visual scenes," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 13 540–13 549.

[104] Y. Mao, J. Zhang, M. Xiang, Y. Zhong, and Y. Dai, "Multimodal variational auto-encoder based audio-visual segmentation," in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2023, pp. 954–965.

[105] J. Liu, Y. Liu, F. Zhang, C. Ju, Y. Zhang, and Y. Wang, "Audio-visual segmentation via unlabeled frame exploitation," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 26 318–26 329.

[106] S. Huang, R. Ling, T. Hui, H. Li, X. Zhou, S. Zhang, S. Liu, R. Hong, and M. Wang, "Revisiting audio-visual segmentation with vision-centric transformer," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 8352–8361.

[107] C. Liu, L. Yang, P. Li, D. Wang, L. Li, and X. Yu, "Dynamic derivation and elimination: Audio visual segmentation with enhanced audio semantics," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 3131–3141.

[108] C. Liu, P. P. Li, Q. Yu, H. Sheng, D. Wang, L. Li, and X. Yu, "Benchmarking audio visual segmentation for long-untrimmed videos," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 22 712–22 722.

[109] Y. Chen, Y. Liu, H. Wang, F. Liu, C. Wang, H. Frazer, and G. Carneiro, "Unraveling instance associations: A closer look for audio-visual segmentation," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 26 487–26 497.

[110] Y. Wang, P. Sun, D. Zhou, G. Li, H. Zhang, and D. Hu, "Ref-avs: Refer and segment objects in audio-visual scenes," in *European conference on computer vision*. Springer, 2024, pp. 196–213.

[111] R. Guo, X. Ying, Y. Chen, D. Niu, G. Li, L. Qu, Y. Qi, J. Zhou, B. Xing, W. Yue *et al.*, "Audio-visual instance segmentation," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 13 550–13 560.

[112] Z. Luo, N. Liu, X. Yang, S. Khan, R. M. Anwer, H. Cholakkal, F. S. Khan, and J. Han, "Tavis: Text-bridged audio-visual segmentation with foundation models," in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 24 014–24 023.

[113] K. Ying, H. Ding, G. Jie, and Y.-G. Jiang, "Towards omnimodal expressions and reasoning in referring audio-visual segmentation," in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 22 575–22 585.

[114] Y. Wu, L. Zhu, Y. Yan, and Y. Yang, "Dual attention matching for audio-visual event localization," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 6292–6300.

[115] H. Xuan, Z. Zhang, S. Chen, J. Yang, and Y. Yan, "Cross-modal attention network for temporal inconsistent audio-visual event localization," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 1, 2020, pp. 279–286.

[116] Y.-B. Lin and Y.-C. F. Wang, "Audiovisual transformer with instance attention for audio-visual event localization," in *Computer Vision – ACCV 2020*. Springer, 2020, pp. 274–290.

[117] Y. Xia and Z. Zhao, "Cross-modal background suppression for audio-visual event localization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 19 989–19 998.

[118] V. R. Rao, M. I. Khalil, H. Li, P. Dai, and J. Lu, "Dual perspective network for audio-visual event localization," in *Computer Vision – ECCV 2022*. Springer, 2022, pp. 689–704.

[119] Y. Wu and Y. Yang, "Exploring heterogeneous clues for weakly-supervised audio-visual video parsing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 1326–1335.

[120] H. Cheng, Z. Liu, H. Zhou, C. Qian, W. Wu, and L. Wang, "Joint-modal label denoising for weakly-supervised audio-visual video parsing," in *Computer Vision – ECCV 2022*. Springer, 2022, pp. 431–448.

[121] Y. Xu, C. Hu, and G. H. Lee, "Rethink cross-modal fusion in weakly-supervised audio-visual video parsing," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 5615–5624.

[122] K. K. Rachavarapu and A. N. Rajagopalan, "Boosting positive segments for weakly-supervised audio-visual video parsing," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10 192–10 202.

[123] K. K. Rachavarapu, K. Ramakrishnan, and A. N. Rajagopalan, "Weakly-supervised audio-visual video parsing with prototype-based pseudo-labeling," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 18 952–18 962.

[124] Y.-H. Lai, J. Ebbers, Y.-C. F. Wang, F. Germain, M. J. Jones, and M. Chatterjee, "UWAV: Uncertainty-weighted weakly-supervised audio-visual video parsing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 13 561–13 570.

[125] L. Wang, B. Zhu, Y. Chen, Y. Zhang, M. Tang, and J. Wang, "MUG: Pseudo labeling augmented audio-visual mamba network for audio-visual video parsing," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 20 637–20 646.

[126] P. Bao, W. Yang, B. P. Ng, M. H. Er, and A. C. Kot, "Cross-modal label contrastive learning for unsupervised audio-visual event localization," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 1, 2023, pp. 215–222.

[127] T. Geng, T. Wang, J. Duan, R. Cong, and F. Zheng, "Dense-localizing audio-visual events in untrimmed videos: A large-scale benchmark and baseline," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 22 942–22 951.

[128] J. Yu, B. Zhang, Z. Teng, and J. Fan, "OpenAVE: Moving towards open set audio-visual event localization," in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 7503–7512.

[129] J. Zhou, D. Guo, R. Guo, Y. Mao, J. Hu, Y. Zhong, X. Chang, and M. Wang, "Towards open-vocabulary audio-visual event localization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 8362–8371.

[130] L. Liu, S. Li, and Y. Zhu, "Audio-visual semantic graph network for audio-visual event localization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 23 957–23 966.

[131] C. Sun, J. Zhang, C. Zhu, M. Huang, and B. Du, "Background-compensated audio-visual semantic modulation framework for audio-visual event localization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings*, 2026, pp. 7272–7281.

[132] A. Ephrat, I. Mosseri, O. Lang, T. Dekel, K. Wilson, A. Hassidim, W. T. Freeman, and M. Rubinstein, "Looking to listen at the cocktail party: A speaker-independent audio-visual model for speech separation," *ACM Transactions on Graphics*, vol. 37, no. 4, pp. 1–11, 2018.

[133] E. Kazakos, A. Nagrani, A. Zisserman, and D. Damen, "EPIC-Fusion: Audio-visual temporal binding for egocentric action recognition," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 5492–5501.

[134] H. M. Fayek and A. Kumar, "Large scale audiovisual learning of sounds with weakly labeled data," in *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, 2020, pp. 558–565.

[135] M. Brousmiche, J. Rouat, and S. Dupont, "Multimodal attentive fusion network for audio-visual event recognition," *Information Fusion*, vol. 85, pp. 52–59, 2022.

[136] J. Li, K. Ma, S. Qu, P.-Y. Huang, and F. Metze, "Audio-visual event recognition through the lens of adversary," in *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 616–620.

[137] W. Wang, D. Tran, and M. Feiszli, "What makes training multi-modal classification networks hard?" in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 12 695–12 705.

[138] C. Huang, Y. Wei, Z. Yang, and D. Hu, "Adaptive unimodal regulation for balanced multimodal information acquisition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 25 854–25 863.

[139] M. Rakib and A. Bagavathi, "G2D: Boosting multimodal learning with gradient-guided distillation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 4059–4068.
[140] Y. Gong, A. H. Liu, A. Rouditchenko, and J. Glass, "UAVM: Towards unifying audio and visual models," *IEEE Signal Processing Letters*, vol. 29, pp. 2437–2441, 2022.
[141] Y.-B. Lin and G. Bertasius, "Siamese vision transformers are scalable audio-visual learners," in *Computer Vision – ECCV 2024*. Springer, 2024, pp. 303–321.
[142] Y. Ishikawa, S. Nakada, H. Munakata, K. Saito, T. Komatsu, and Y. Aoki, "Language-guided contrastive audio-visual masked autoencoder with automatically generated audio-visual-text triplets from videos," in *Proceedings of Interspeech 2025*, 2025, pp. 2605–2609.
[143] R. Gao, R. Feris, and K. Grauman, "Learning to separate object sounds by watching unlabeled video," in *Proceedings of the European Conference on Computer Vision*, 2018, pp. 35–53.
[144] R. Gao and K. Grauman, "Co-separating sounds of visual objects," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 3879–3888.
[145] H. Zhao, C. Gan, W.-C. Ma, and A. Torralba, "The sound of motions," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 1735–1744.
[146] C. Gan, D. Huang, H. Zhao, J. B. Tenenbaum, and A. Torralba, "Music gesture for visual sound separation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 10 478–10 487.
[147] L. Zhu and E. Rahtu, "Visually guided sound source separation using cascaded opponent filter network," in *Proceedings of the Asian Conference on Computer Vision*, 2020, pp. 409–426.
[148] Y. Tian, D. Hu, and C. Xu, "Cyclic co-learning of sounding object visual grounding and sound separation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 2745–2754.
[149] L. Zhu and E. Rahtu, "Visually guided sound source separation and localization using self-supervised motion representations," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 1289–1299.
[150] ——, "V-SlowFast network for efficient visual sound separation," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 1464–1474.
[151] C. Li and Y. Qian, "Deep audio-visual speech separation with attention mechanism," in *2020 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2020, pp. 7314–7318.
[152] S.-W. Chung, S. Choe, J. S. Chung, and H.-G. Kang, "FaceFilter: Audio-visual speech separation using still images," in *Proceedings of Interspeech 2020*, 2020, pp. 3481–3485.
[153] R. Gao and K. Grauman, "VisualVoice: Audio-visual speech separation with cross-modal consistency," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 15 495–15 505.
[154] J. Chen, R. Zhang, D. Lian, J. Yang, Z. Zeng, and J. Shi, "iQuery: Instruments as queries for audio-visual sound separation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 14 675–14 686.
[155] Y. Su, A. Vosoughi, S. Deng, Y. Tian, and C. Xu, "Separating invisible sounds toward universal audio-visual scene-aware sound separation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2023.
[156] Y. Ye, W. Yang, and Y. Tian, "LAVSS: Location-guided audio-visual spatial audio separation," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 5508–5519.
[157] M. A. Islam, S. S. Nabavi, I. Kezele, Y. Wang, Y. Yu, and J. Tang, "Visually guided audio source separation with meta consistency learning," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 3014–3023.
[158] W. Pian, Y. Nan, S. Deng, S. Mo, Y. Guo, and Y. Tian, "Continual audio-visual sound separation," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 76 058–76 079.
[159] C. Huang, S. Liang, Y. Tian, A. Kumar, and C. Xu, "High-quality visually-guided sound separation from diverse categories," in *Proceedings of the Asian Conference on Computer Vision*, 2024, pp. 35–49.
[160] Y. Yu and S. Sun, "DGFNet: End-to-end audio-visual source separation based on dynamic gating fusion," in *Proceedings of the 2025 International Conference on Multimedia Retrieval*, 2025, pp. 1730–1738.
[161] H. Hu, D. Lin, Q. Huang, Y. Hou, H. J. Chang, and J. Jiao, "Audio-visual separation with hierarchical fusion and representation alignment," in *Proceedings of the British Machine Vision Conference*. BMVA, 2025.
[162] D. Zhang, Y. Deng, Y. Wei, and B. Ji, "AVFSNet: Audio-visual speech separation for flexible number of speakers with multi-scale and multi-task learning," *Speech Communication*, vol. 179, p. 103378, 2026.
[163] E. Kazakos, A. Nagrani, A. Zisserman, and D. Damen, "Slow-fast auditory streams for audio recognition," in *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 855–859.
[164] M. Subedar, R. Krishnan, P. L. Meyer, O. Tickoo, and J. Huang, "Uncertainty-aware audiovisual activity recognition using deep bayesian variational inference," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 6301–6310.
[165] R. Gao, T.-H. Oh, K. Grauman, and L. Torresani, "Listen to look: Action recognition by previewing audio," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 10 457–10 467.
[166] R. Panda, C.-F. Chen, Q. Fan, X. Sun, K. Saenko, A. Oliva, and R. Feris, "AdaMML: Adaptive multi-modal learning for efficient video recognition," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 7576–7585.
[167] Y. Liu, Y. Tan, and H. Lan, "Self-supervised contrastive learning for audio-visual action recognition," in *2023 IEEE International Conference on Image Processing (ICIP)*, 2023, pp. 1000–1004.
[168] C. Chen, K. Ashutosh, R. Girdhar, D. Harwath, and K. Grauman, "SoundingActions: Learning how actions sound from narrated egocentric videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 252–27 262.
[169] J. Chalk, J. Huh, E. Kazakos, A. Zisserman, and D. Damen, "TIM: A time interval machine for audio-visual action recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 18 153–18 163.
[170] S. Huang, J. Wu, X. Wei, Y. Cai, D. Jiang, and Y. Wang, "Sound bridge: Associating egocentric and exocentric videos via audio cues," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 28 942–28 951.
[171] J. Lee, J. Chang, D. Lee, and J. Choi, "CA^{2}ST: Cross-attention in audio, space, and time for holistic video recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 48, no. 3, pp. 2803–2819, 2026.
[172] M. Planamente, C. Plizzari, E. Alberti, and B. Caputo, "Domain generalization through audio-visual relative norm alignment in first person action recognition," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 1807–1818.
[173] Y. Zhang, H. Doughty, L. Shao, and C. G. M. Snoek, "Audio-adaptive activity recognition across video domains," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 13 791–13 800.
[174] W. Pian, S. Mo, Y. Guo, and Y. Tian, "Audio-visual class-incremental learning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 7799–7811.
[175] Y. Wei and D. Hu, "MMPareto: Boosting multimodal learning with innocent unimodal assistance," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 52 559–52 572.
[176] S. Chowdhury, S. Biswas, S. Nag, T. Nagarajan, C. Murdock, I. Ananthabhotla, Y. Qian, V. K. Ithapu, D. Manocha, and R. Gao, "EgoAdapt: Adaptive multisensory distillation and policy learning for efficient egocentric perception," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 10 741–10 752.
[177] D. Zverev, T. Wiedemer, A. Prabhu, M. Bethge, W. Brendel, and A. S. Koepke, "VGGSounder: Audio-visual evaluations for foundation models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 1027–1037.
[178] J. Huh, J. Chalk, E. Kazakos, D. Damen, and A. Zisserman, "EPIC-SOUNDS: A large-scale dataset of actions that sound," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 11, pp. 9953–9965, 2025.
[179] T. Perrett, A. Darkhalil, S. Sinha, O. Emara, S. Pollard, K. K. Parida, K. Liu, P. Gatti, S. Bansal, K. Flanagan, J. Chalk, Z. Zhu, R. Guerrier, F. Abdelazim, B. Zhu, D. Moltisanti, M. Wray, H. Doughty, and D. Damen, "HD-EPIC: A highly-detailed egocentric video dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 23 901–23 913.
[180] H. Bello, L. Ray, J. Sorysz, S. Suh, and P. Lukowicz, "OpenMarcie: Dataset for multimodal action recognition in industrial environments,"

in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 20 129–20 138.

[181] M. W. Hussain, J. Civera, and L. Montano, "Grounding acoustic echoes in single view geometry estimation," in *Proceedings of the Twenty-Eighth AAAI Conference on Artificial Intelligence*, 2014, pp. 2760–2766.

[182] R. Gao, C. Chen, Z. Al-Halah, C. Schissler, and K. Grauman, "VisualEchoes: Spatial image representation learning through echolocation," in *Computer Vision – ECCV 2020*. Springer, 2020, pp. 658–676.

[183] N. Singh, J. Mentch, J. Ng, M. Beveridge, and I. Drori, "Image2Reverb: Cross-modal reverb impulse response synthesis," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 286–295.

[184] A. Luo, Y. Du, M. J. Tarr, J. B. Tenenbaum, A. Torralba, and C. Gan, "Learning neural acoustic fields," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 3165–3177.

[185] J. H. Christensen, S. Hornauer, and S. X. Yu, "BatVision: Learning to see 3d spatial layout with two ears," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 1581–1587.

[186] K. K. Parida, S. Srivastava, and G. Sharma, "Beyond image to depth: Improving depth prediction using echoes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 8268–8277.

[187] G. Irie, T. Shibata, and A. Kimura, "Co-attention-guided bilinear model for echo-based depth estimation," in *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 4648–4652.

[188] X. Liu, S. Hornauer, F. Moutarde, and J. Lu, "AVS-Net: Audio-visual scale net for self-supervised monocular metric depth estimation," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, Sight and Sound Workshop*, 2025, cVPR 2025 Sight and Sound Workshop.

[189] G. Liu, W. Cui, Y. Xi, L. Yang, P. Hu, H. Kong, and Z. Wang, "SAGENet: Binaural echo-based 3d depth estimation with sparse angular queries and refined geometric cues," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 6113–6120.

[190] S. Purushwalkam, S. V. Amengual Garí, V. K. Ithapu, C. Schissler, P. Robinson, A. Gupta, and K. Grauman, "Audio-visual floorplan reconstruction," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 1183–1192.

[191] X. Hu, S. Purushwalkam, D. Harwath, and K. Grauman, "Learning to map efficiently by active echolocation," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 1505–1510.

[192] W. Luo, Q. Song, Z. Yan, R. Tan, and G. Lin, "Indoor smartphone SLAM with acoustic echoes," *IEEE Transactions on Mobile Computing*, vol. 23, no. 6, pp. 6634–6649, 2024.

[193] Z. Yang, X. Fan, V. Isler, and H. S. Park, "PoseKernelLifter: Metric lifting of 3d human pose using sound," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 13 179–13 189.

[194] Y. Shibata, Y. Kawashima, M. Isogawa, G. Irie, A. Kimura, and Y. Aoki, "Listening human behavior: 3d human pose estimation with acoustic signals," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 13 323–13 332.

[195] Y. Oumi, Y. Shibata, G. Irie, A. Kimura, Y. Aoki, and M. Isogawa, "Acoustic-based 3d human pose estimation robust to human position," in *Proceedings of the British Machine Vision Conference*. BMVA, 2024.

[196] Y. Shibata, Y. Oumi, G. Irie, A. Kimura, Y. Aoki, and M. Isogawa, "BGM2Pose: Active 3d human pose estimation with non-stationary sounds," *IEEE Open Journal of Signal Processing*, vol. 7, pp. 790–799, 2026.

[197] Y. Oumi, Y. Shibata, G. Irie, A. Kimura, Y. Aoki, and M. Isogawa, "Sound-based multi-person 3d pose estimation," in *Proceedings of the European Conference on Computer Vision*, 2026.

[198] D. Li, T. R. Langlois, and C. Zheng, "Scene-aware audio for 360° videos," *ACM Transactions on Graphics*, vol. 37, no. 4, pp. 111:1–111:12, 2018.

[199] C. Chen, R. Gao, P. Calamia, and K. Grauman, "Visual acoustic matching," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 18 858–18 868.

[200] A. Ratnarajah, S. Ghosh, S. Kumar, P. Chiniya, and D. Manocha, "AV-RIR: Audio-visual room impulse response estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 164–27 175.

[201] M. F. Saad and Z. Al-Halah, "How would it sound? material-controlled multimodal acoustic profile generation for indoor scenes," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 12 232–12 241.

[202] M. F. Saad, S. Majumder, K. Grauman, and Z. Al-Halah, "Materialistic RIR: Material conditioned realistic RIR generation," 2026, accepted to CVPR 2026 Findings.

[203] K. Su, M. Chen, and E. Shlizerman, "INRAS: Implicit neural representation for audio scenes," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 8144–8158.

[204] S. Liang, C. Huang, Y. Tian, A. Kumar, and C. Xu, "Neural acoustic context field: Rendering realistic room impulse response with neural fields," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2023, workshop on Audio-Visual Learning for 4D Scene Understanding.

[205] ——, "AV-NeRF: Learning neural fields for real-world audio-visual scene synthesis," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 37 472–37 490.

[206] S. Bhosale, H. Yang, D. Kanojia, J. Deng, and X. Zhu, "AV-GS: Learning material and geometry aware priors for novel view acoustic synthesis," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[207] A. Brunetto, S. Hornauer, and F. Moutarde, "NeRAF: 3d scene infused neural radiance and acoustic fields," in *International Conference on Learning Representations*, 2025.

[208] D. Jin and R. Gao, "Differentiable room acoustic rendering with multi-view vision priors," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 37–47.

[209] S. Majumder, C. Chen, Z. Al-Halah, and K. Grauman, "Few-shot audio-visual learning of environment acoustics," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 2522–2536.

[210] C. Chen, A. Richard, R. Shapovalov, V. K. Ithapu, N. Neverova, K. Grauman, and A. Vedaldi, "Novel-view acoustic synthesis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 6409–6419.

[211] M. Chen, K. Su, and E. Shlizerman, "Be everywhere - hear everything (BEE): Audio scene reconstruction by sparse audio-visual samples," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 7853–7862.

[212] M. Chen and E. Shlizerman, "AV-Cloud: Spatial audio rendering through audio-visual cloud splatting," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 141 021–141 044.

[213] M. Chen, I. D. Gebru, I. Ananthabhotla, C. Richardt, D. Markovic, J. Sandakly, S. Krenn, T. Keebler, E. Shlizerman, and A. Richard, "SoundVista: Novel-view ambient sound synthesis via visual-acoustic binding," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 8331–8341.

[214] H. Gao, J. Ma, D. Ahmedt-Aristizabal, C. Nguyen, and M. Liu, "SOAF: Scene occlusion-aware neural acoustic field," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2026, pp. 4097–4107.

[215] A. Somayazulu, S. Majumder, C. Chen, Z. Al-Halah, and K. Grauman, "Sample-efficient audio-visual learning of scene acoustics," *International Journal of Computer Vision*, vol. 134, p. 193, 2026, article 193.

[216] D. Jin, X. Chen, M. C. Lin, and R. Gao, "SonoWorld: From one image to a 3d audio-visual scene," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 30 194–30 204.

[217] A. Brunetto, S. Hornauer, S. X. Yu, and F. Moutarde, "The audio-visual BatVision dataset for research on sight and sound," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 5812–5819.

[218] Z. Chen, I. D. Gebru, C. Richardt, A. Kumar, W. Laney, A. Owens, and A. Richard, "Real acoustic fields: An audio-visual room acoustics dataset and benchmark," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 21 886–21 896.

[219] P. Yang, X. Wang, X. Duan, H. Chen, R. Hou, C. Jin, and W. Zhu, "AVQA: A dataset for audio-visual question answering on videos," in *Proceedings of the 30th ACM International Conference on Multimedia*, 2022, pp. 3480–3491. [Online]. Available: https://doi.org/10.1145/3503161.3548291

[220] G. Li, W. Hou, and D. Hu, "Progressive spatio-temporal perception for audio-visual question answering," in *Proceedings of the 31st ACM International Conference on Multimedia*, 2023, pp. 7808–7816. [Online]. Available: https://doi.org/10.1145/3581783.3612293

[221] A. Nadeem, A. Hilton, R. Dawes, G. Thomas, and A. Mustafa, "CAD: Contextual multi-modal alignment for dynamic AVQA,"

in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024. [Online]. Available: https://openaccess.thecvf.com/content/WACV2024/html/Nadeem_CAD_-_Contextual_Multi-Modal_Alignment_for_Dynamic_AVQA_WACV_2024_paper.html

[222] H. Kim, I. Jung, D. Suh, Y. Zhang, S. Lee, and S. Hong, "Question-aware gaussian experts for audio-visual question answering," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 13 681–13 690. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Kim_Question-Aware_Gaussian_Experts_for_Audio-Visual_Question_Answering_CVPR_2025_paper.html

[223] Y.-B. Lin, Y.-L. Sung, J. Lei, M. Bansal, and G. Bertasius, "Vision transformers are parameter-efficient audio-visual learners," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 2299–2309.

[224] H. Duan, Y. Xia, M. Zhou, L. Tang, J. Zhu, and Z. Zhao, "Cross-modal prompts: Adapting large pre-trained models for audio-visual downstream tasks," in *Advances in Neural Information Processing Systems*, 2023. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/af01716e08073368a7c8a62be46dba17-Abstract.html

[225] X. Liu, Z. Dong, and P. Zhang, "Tackling data bias in MUSIC-AVQA: Crafting a balanced dataset for unbiased question-answering," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 4478–4487. [Online]. Available: https://openaccess.thecvf.com/content/WACV2024/html/Liu_Tackling_Data_Bias_in_MUSIC-AVQA_Crafting_a_Balanced_Dataset_for_WACV_2024_paper.html

[226] J. Ma, M. Hu, P. Wang, W. Sun, L. Song, H. Pei, J. Liu, and Y. Du, "Look, listen, and answer: Overcoming biases for audio-visual question answering," in *Advances in Neural Information Processing Systems*, vol. 37, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/123a18dfd821c8b440f42a00a27648d6-Abstract-Conference.html

[227] K. Wu, X. Li, X. Li, C. Hu, and G. Wu, "AVQACL: A novel benchmark for audio-visual question answering continual learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 3252–3261. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Wu_AVQACL_A_Novel_Benchmark_for_Audio-Visual_Question_Answering_Continual_Learning_CVPR_2025_paper.html

[228] S. Majumder, H. Jiang, P. Moulon, E. Henderson, P. Calamia, K. Grauman, and V. K. Ithapu, "Chat2Map: Efficient scene mapping from multi-ego conversations," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 10 554–10 564. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2023/html/Majumder_Chat2Map_Efficient_Scene_Mapping_From_Multi-Ego_Conversations_CVPR_2023_paper.html

[229] M. Chen, Z. Cui, X. Liu, J. Xiang, Y. Zheng, J. Li, and E. Shlizerman, "SAVVY: Spatial awareness via audio-visual LLMs through seeing and hearing," in *Advances in Neural Information Processing Systems*, vol. 38, 2025. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/file/ac895e51849bfc99ae25e054fd4c2eda-Paper-Conference.pdf

[230] H. Ryu, J. S. Chung, and D. Harwath, "Hear you are: Teaching LLMs spatial reasoning with vision and spatial sound," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/html/Ryu_Hear_you_are_Teaching_LLMs_Spatial_Reasoning_with_Vision_and_CVPR_2026_paper.html

[231] Z. Liu, C. Tang, Y. Wang, Z. Zhu, Y. Chen, Y. Shao, T. Wang, L. Ke, Z. Jin, and C. Zhang, "JAEGER: Joint 3D audio-visual grounding and reasoning in simulated physical environments," *arXiv preprint arXiv:2602.18527*, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2602.18527

[232] M. Chen, Z. Cui, R. Zhang, H. Ryu, and E. Shlizerman, "SceneBind: Binding what and where across vision, audio and language," *arXiv preprint arXiv:2607.15265*, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2607.15265

[233] L. Zhu, T. Geng, Y. Chen, T. Wang, P. Lu, and F. Zheng, "R-AVST: Empowering video-LLMs with fine-grained spatio-temporal reasoning in complex audio-visual scenarios," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 9, 2026, pp. 7627–7635. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/37704

[234] Z. Zeng, C. Zhang, Z. Yang, R. Pi, J. Wu, D. Zhang, Z. Ma, G. Li, Z. Yang, Y. Xiang, Y. Zheng, and M. Luo, "Listen, see and track: Spatio-temporal audio-visual sound event reasoning for omni-modal language models," *arXiv preprint arXiv:2608.09435*, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2608.09435

[235] B. Zhu, Y. Fu, Q. Dong, G. Sun, T. Qian, Y. Wu, D. P. Paudel, Y. Fu, and X. Xue, "EgoSound: Benchmarking sound understanding in egocentric videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/html/Zhu_EgoSound_Benchmarking_Sound_Understanding_in_Egocentric_Videos_CVPR_2026_paper.html

[236] C. Lv, S. Zhang, Y. Tian, M. Qi, and H. Ma, "Disentangled counterfactual learning for physical audiovisual commonsense reasoning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 12 476–12 488. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/29571f8fda54fe93631c41aad4215abc-Abstract-Conference.html

[237] D. Zong, C. Ding, K. Chen, Y. Li, and S. Wang, "Counterfactual debiasing for physical audiovisual commonsense reasoning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 14, 2025, pp. 15 265–15 273. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/33675

[238] M. Qi, C. Lv, and H. Ma, "Robust disentangled counterfactual learning for physical audiovisual commonsense reasoning," *arXiv preprint arXiv:2502.12425*, 2025, preprint. [Online]. Available: https://arxiv.org/abs/2502.12425

[239] W. Chow, J. Mao, B. Li, D. Seita, V. C. Guizilini, and Y. Wang, "Physbench: Benchmarking and enhancing vision-language models for physical world understanding," in *International Conference on Learning Representations*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/f38cb4cf9a5eaa92b3cfa481832719c6-Abstract-Conference.html

[240] A. Foss, C. Evans, S. Mitts, K. Sinha, A. Rizvi, and J. T. Kao, "Causalvqa: A physically grounded causal reasoning benchmark for video models," *arXiv preprint arXiv:2506.09943*, 2025, preprint. [Online]. Available: https://arxiv.org/abs/2506.09943

[241] B. Krojer, M. Komeili, C. Ross, Q. Garrido, K. Sinha, N. Ballas, and M. Assran, "A shortcut-aware video-qa benchmark for physical understanding via minimal video pairs," *Transactions on Machine Learning Research*, 2025. [Online]. Available: https://openreview.net/forum?id=gvFgNJcSw1

[242] H.-B. Oh, Y. Takida, T. Uesaka, T.-H. Oh, and Y. Mitsufuji, "Pavas: Physics-aware video-to-audio synthesis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 14 481–14 491. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/html/Hyun-Bin_PAVAS_Physics-Aware_Video-to-Audio_Synthesis_CVPR_2026_paper.html

[243] T. Li, S. Gururani, K. J. Shih, G. Bhatt, S.-g. Lee, Z. Kong, A. Goel, G. Anumanchipalli, and M.-Y. Liu, "Benchmarking single-factor physical video-to-audio generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 1939–1949. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/html/Li_Benchmarking_Single-Factor_Physical_Video-to-Audio_Generation_CVPR_2026_paper.html

[244] G. Sun, W. Yu, C. Tang, X. Chen, T. Tan, W. Li, L. Lu, Z. Ma, Y. Wang, and C. Zhang, "video-SALMONN: Speech-enhanced audio-visual large language models," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 47 198–47 217. [Online]. Available: https://proceedings.mlr.press/v235/sun24l.html

[245] G. Sun, Y. Yang, J. Zhuang, C. Tang, Y. Li, W. Li, Z. Ma, and C. Zhang, "video-SALMONN-o1: Reasoning-enhanced audio-visual large language model," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 57 808–57 827. [Online]. Available: https://proceedings.mlr.press/v267/sun25x.html

[246] T. Geng, J. Zhang, Q. Wang, T. Wang, J. Duan, and F. Zheng, "LongVALE: Vision-audio-language-event benchmark towards time-aware omni-modal perception of long videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 18 959–18 969. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Geng_LongVALE_Vision-Audio-Language-Event_Benchmark_Towards_Time-Aware_Omni-Modal_Perception_of_Long_Videos_CVPR_2025_paper.html

[247] Y. Su, T. Lan, H. Li, J. Xu, Y. Wang, and D. Cai, "PandaGPT: One model to instruction-follow them all," in *Proceedings of the 1st*

Workshop on Taming Large Language Models: Controllability in the era of Interactive Assistants, 2023, pp. 11–23. [Online]. Available: https://aclanthology.org/2023.tllm-1.2/

[248] H. Zhang, X. Li, and L. Bing, "Video-LLaMA: An instruction-tuned audio-visual language model for video understanding," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*. Association for Computational Linguistics, 2023, pp. 543–553. [Online]. Available: https://aclanthology.org/2023.emnlp-demo.49/

[249] J. Han, K. Gong, Y. Zhang, J. Wang, K. Zhang, D. Lin, Y. Qiao, P. Gao, and X. Yue, "OneLLM: One framework to align all modalities with language," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 26 584–26 595. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Han_OneLLM_One_Framework_to_Align_All_Modalities_with_Language_CVPR_2024_paper.html

[250] Z. Cheng, S. Leng, H. Zhang, Y. Xin, X. Li, G. Chen, Y. Zhu, W. Zhang, Z. Luo, D. Zhao, and L. Bing, "VideoLLaMA 2: Advancing spatial-temporal modeling and audio understanding in video-LLMs," *arXiv preprint arXiv:2406.07476*, 2024. [Online]. Available: https://arxiv.org/abs/2406.07476

[251] S. Yu, J. Yoon, and M. Bansal, "CREMA: Generalizable and efficient video-language reasoning via multimodal modular fusion," in *International Conference on Learning Representations*, 2025. [Online]. Available: https://arxiv.org/abs/2402.05889

[252] C. Fu, H. Lin, Z. Long, Y. Shen, M. Zhao, Y. Zhang, S. Dong, X. Wang, D. Yin, L. Ma, X. Zheng, R. He, R. Ji, Y. Wu, C. Shan, and X. Sun, "VITA: Towards open-source interactive omni multimodal LLM," *arXiv preprint arXiv:2408.05211*, 2024. [Online]. Available: https://arxiv.org/abs/2408.05211

[253] Z. Liu, Y. Dong, J. Wang, Z. Liu, W. Hu, J. Lu, and Y. Rao, "Ola: Pushing the frontiers of omni-modal language model," *arXiv preprint arXiv:2502.04328*, 2025. [Online]. Available: https://arxiv.org/abs/2502.04328

[254] J. Xu, Z. Guo, J. He, H. Hu, T. He, S. Bai, K. Chen, J. Wang, Y. Fan, K. Dang, B. Zhang, X. Wang, Y. Chu, and J. Lin, "Qwen2.5-Omni technical report," *arXiv preprint arXiv:2503.20215*, 2025.

[255] J. Xu, Z. Guo, H. Hu, Y. Chu, X. Wang, J. He, Y. Wang, X. Shi, T. He, X. Zhu, Y. Lv, Y. Wang, D. Guo, H. Wang, L. Ma, P. Zhang, X. Zhang, H. Hao, Z. Guo, B. Yang, B. Zhang, Z. Ma, X. Wei, S. Bai, K. Chen, X. Liu, P. Wang, M. Yang, D. Liu, X. Ren, B. Zheng, R. Men, F. Zhou, B. Yu, J. Yang, L. Yu, J. Zhou, and J. Lin, "Qwen3-Omni technical report," *arXiv preprint arXiv:2509.17765*, 2025.

[256] Y. Li, G. Zhang, Y. Ma, R. Yuan, K. Zhu, H. Guo, Y. Liang, J. Liu, Z. Wang, J. Yang, S. Wu, X. Qu, J. Shi, X. Zhang, Z. Yang, X. Wang, Z. Zhang, Z. Liu, E. Benetos, W. Huang, and C. Lin, "OmniBench: Towards the future of universal omni-language models," *arXiv preprint arXiv:2409.15272*, 2024, preprint. [Online]. Available: https://arxiv.org/abs/2409.15272

[257] J. Hong, S. Yan, J. Cai, X. Jiang, Y. Hu, and W. Xie, "WorldSense: Evaluating real-world omnimodal understanding for multimodal LLMs," *arXiv preprint arXiv:2502.04326*, 2025, preprint. [Online]. Available: https://arxiv.org/abs/2502.04326

[258] K. Gong, K. Feng, B. Li, Y. Wang, M. Cheng, S. Yang, J. Han, B. Wang, Y. Bai, Z. Yang, and X. Yue, "AV-Odyssey Bench: Can your multimodal LLMs really understand audio-visual information?" *arXiv preprint arXiv:2412.02611*, 2024, preprint. [Online]. Available: https://arxiv.org/abs/2412.02611

[259] Z. Zhou, R. Wang, Z. Wu, and Y.-G. Jiang, "Daily-Omni: Towards audio-visual reasoning with temporal alignment across modalities," 2026.

[260] S. Chowdhury, H. Gani, N. Anand, S. Nag, R. Gao, M. Elhoseiny, S. Khan, and D. Manocha, "AURELIA: Test-time reasoning distillation in audio-visual LLMs," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025. [Online]. Available: https://arxiv.org/abs/2503.23219

[261] Y. Kulkarni and P. Fazli, "AVATAR: Reinforcement learning to see, hear, and reason over video," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/html/Kulkarni_AVATAR_Reinforcement_Learning_to_See_Hear_and_Reason_Over_Video_CVPR_2026_paper.html

[262] Z. Chen, J. Tao, R. Li, Y. Hu, R. Chen, Z. Yang, X. Yu, H. Jing, M. Zhang, S. Shao, B. Wang, Q. Lu, and R. Huang, "OmniVideo-R1: Reinforcing audio-visual reasoning with query intention and modality attention," *arXiv preprint arXiv:2602.05847*, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2602.05847

[263] Y. Dai, Z. Wu, B. Zeng, D. Hua, J. Liu, B. Li, Y. Wang, C. Tong, H. Liang, X. Ma, J. Niu, T. Guo, Y. Shi, Y. Ding, Y. Ji, B. Mei, Y. Guan, Y. Zhang, P. Wan, F. Fu, and W. Zhang, "LatentOmni: Rethinking omni-modal understanding via unified audio-visual latent reasoning," *arXiv preprint arXiv:2605.22012*, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2605.22012

[264] A. Chaubey, J. Pang, M. Siniukov, and M. Soleymani, "AVERE: Improving audiovisual emotion reasoning with preference optimization," in *International Conference on Learning Representations*, 2026. [Online]. Available: https://openreview.net/forum?id=td682AAuPr

[265] C. Fu, Y. Dai, Y. Luo, L. Li, S. Ren, R. Zhang, Z. Wang, C. Zhou, Y. Shen, M. Zhang, P. Chen, Y. Li, S. Lin, S. Zhao, K. Li, T. Xu, X. Zheng, E. Chen, C. Shan, R. He, and X. Sun, "Video-MME: The first-ever comprehensive evaluation benchmark of multi-modal LLMs in video analysis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 24 108–24 118. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Fu_Video-MME_The_First-Ever_Comprehensive_Evaluation_Benchmark_of_Multi-modal_LLMs_in_CVPR_2025_paper.html

[266] Z. Li, X. Zhang, Y. Guo, M. Bennamoun, F. Boussaid, G. Dwivedi, L. Gong, and Q. Ke, "Watch and listen: Understanding audio-visual-speech moments with multimodal LLM," in *Advances in Neural Information Processing Systems*, vol. 38, 2025. [Online]. Available: https://papers.nips.cc/paper_files/paper/2025/file/bbe0acd1ad5cf41bb87a484992a70cbd-Paper-Conference.pdf

[267] G. Sun, Y. Li, X. Wu, Y. Yang, W. Li, Z. Ma, and C. Zhang, "video-SALMONN S: Memory-enhanced streaming audio-visual LLM," *arXiv preprint arXiv:2510.11129*, 2026, preprint; title updated in version 2. [Online]. Available: https://arxiv.org/abs/2510.11129

[268] G. Sun, Y. Li, Y. Yang, and C. Zhang, "OmniMem: Perturbation-aware memory compression for streaming audio-visual LLMs," *arXiv preprint arXiv:2606.07577*, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2606.07577

[269] K. Tao, Y. Zheng, J. Xu, W. Du, K. Shao, H. Wang, X. Chen, X. Jin, J. Zhu, B. Yu, W. Wang, J. Liu, C. Qin, Y. Zhang, M.-H. Yang, and H. Wang, "LVOmniBench: Pioneering long audio-video understanding evaluation for omnimodal LLMs," *arXiv preprint arXiv:2603.19217*, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2603.19217

[270] K. Tao, W. Du, B. Yu, W. Wang, J. Liu, and H. Wang, "OmniAgent: Audio-guided active perception agent for omnimodal audio-video understanding," *arXiv preprint arXiv:2512.23646*, 2025, preprint. [Online]. Available: https://arxiv.org/abs/2512.23646

[271] J. Yang, S. Liu, H. Guo, Y. Dong, X. Zhang, S. Zhang, P. Wang, Z. Zhou, B. Xie, Z. Wang, B. Ouyang, Z. Lin, M. Cominelli, Z. Cai, B. Li, Y. Zhang, P. Zhang, F. Hong, J. Widmer, F. Gringoli, L. Yang, and Z. Liu, "EgoLife: Towards egocentric life assistant," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 28 885–28 900. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Yang_EgoLife_Towards_Egocentric_Life_Assistant_CVPR_2025_paper.html

[272] S. Chowdhury, M. Elmoghany, Y. Abeysinghe, J. Fei, S. Nag, S. H. Khan, M. Elhoseiny, and D. Manocha, "MAGNET: A multi-agent framework for finding audio-visual needles by reasoning over multi-video haystacks," in *Advances in Neural Information Processing Systems*, vol. 38, 2025, pp. 54 971–55 007. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/465c08c85e6f1015358f2e0c0a0cbaf9-Abstract-Conference.html

[273] S. Mo and Y. Song, "Aligning audio-visual joint representations with an agentic workflow," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 58 841–58 867. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/6c0ff499edc529c7d8c9f05c7c0ccb82-Abstract-Conference.html

[274] Y. Luo, X. Zheng, G. Li, S. Yin, H. Lin, C. Fu, J. Huang, J. Ji, F. Chao, J. Luo, and R. Ji, "Video-RAG: Visually-aligned retrieval-augmented long video comprehension," in *Advances in Neural Information Processing Systems*, 2025. [Online]. Available: https://arxiv.org/abs/2411.13093

[275] D. Galanopoulos, A. Goulas, A. Leventakis, I. Patras, and V. Mezaris, "An LLM framework for long-form video retrieval and audio-visual question answering using Qwen2/2.5," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2025, pp. 3769–3778. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025W/IViSE/html/Galanopoulos_An_LLM_Framework_for_Long-form_Video_Retrieval_and_Audio-Visual_Question_CVPRW_2025_paper.html

[276] Y. Zhu, X. Mu, T. Feng, Z. Ou, Y. Gong, and H. Luo, "OmniRAG-Agent: Agentic omnimodal reasoning for low-resource long audio-

video question answering," *arXiv preprint arXiv:2602.03707*, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2602.03707

[277] Z. Xing, R. Xu, Y. Wang, J. He, Z. Ma, Q. Yang, Y. Chu, J. Xu, J. Lin, C.-W. Fu, and P.-A. Heng, "Native active perception as reasoning for omni-modal understanding," *arXiv preprint arXiv:2606.19341*, 2026. [Online]. Available: https://arxiv.org/abs/2606.19341

[278] Y. Fan, X. Ma, R. Wu, Y. Du, J. Li, Z. Gao, and Q. Li, "VideoAgent: A memory-augmented multimodal agent for video understanding," in *Computer Vision–ECCV 2024*, 2024. [Online]. Available: https://arxiv.org/abs/2403.11481

[279] Y. Lin, J. Zhang, Q. Wang, H. Ye, Y. Fu, Y. Liu, H. H. Li, and Y. Chen, "HippoMM: Hippocampal-inspired multimodal memory for long audiovisual event understanding," *arXiv preprint arXiv:2504.10739*, 2026, current revised version; reported as CVPR 2026 Findings. [Online]. Available: https://arxiv.org/abs/2504.10739

[280] V. Pipoli, A. Saporita, F. Bolelli, M. Cornia, L. Baraldi, C. Grana, R. Cucchiara, and E. Ficarra, "MissRAG: Addressing the missing modality challenge in multimodal large language models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 3215–3224. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025/html/Pipoli_MissRAG_Addressing_the_Missing_Modality_Challenge_in_Multimodal_Large_Language_ICCV_2025_paper.html

[281] Z. Wang, Y. Zhang, S. Yu, C. Zhang, Z. Zhao, J. Yoon, H. Lee, G. Bertasius, and M. Bansal, "EgoMemReason: A memory-driven reasoning benchmark for long-horizon egocentric video understanding," in *Conference on Language Modeling*, 2026, accepted at COLM 2026; proceedings metadata pending. [Online]. Available: https://arxiv.org/abs/2605.09874

[282] Y. Cao, X. Min, Y. Gao, W. Sun, Z. Zhang, J. Han, and G. Zhai, "XGC-AVis: Towards audio-visual content understanding with a multi-agent collaborative system," *arXiv preprint arXiv:2509.23251*, 2025, preprint. [Online]. Available: https://arxiv.org/abs/2509.23251

[283] Y. Yu, W. Huang, F. Sun, C. Chen, Y. Wang, and X. Liu, "Sound adversarial audio-visual navigation," in *International Conference on Learning Representations*, 2022.

[284] H. Kondoh and A. Kanezaki, "Multi-goal audio-visual navigation using sound direction map," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 5219–5226.

[285] S. Paul, A. K. Roy Chowdhury, and A. Cherian, "AVLEN: Audio-visual-language embodied navigation in 3d environments," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 6236–6249.

[286] J. Chen, W. Wang, S. Liu, H. Li, and Y. Yang, "Omnidirectional information gathering for knowledge transfer-based audio-visual navigation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10 993–11 003.

[287] X. Liu, S. Paul, M. Chatterjee, and A. Cherian, "CAVEN: An embodied conversational agent for efficient audio-visual navigation in noisy environments," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 4, 2024, pp. 3765–3773.

[288] Z. Yang, J. Liu, P. Chen, A. Cherian, T. K. Marks, J. Le Roux, and C. Gan, "RILA: Reflective and imaginative language agent for zero-shot semantic audio-visual navigation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16 251–16 261.

[289] Z. Shi, L. Zhang, L. Li, and Y. Shen, "Towards audio-visual navigation in noisy environments: A large-scale benchmark dataset and an architecture considering multiple sound-sources," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 14, 2025, pp. 14 673–14 680.

[290] H. Kondoh and A. Kanezaki, "Embodied navigation with auxiliary task of action description prediction," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 7025–7036.

[291] Y. Zeng, H. Wang, M. Liu, Y. Zhou, C. Gao, K. Chen, and G. Huang, "Semantic audio-visual navigation in continuous environments," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 22 369–22 379.

[292] C. Gan, Y. Gu, S. Zhou, J. Schwartz, S. Alter, J. Traer, D. Gutfreund, J. B. Tenenbaum, J. H. McDermott, and A. Torralba, "Finding fallen objects via asynchronous audio-visual integration," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 10 523–10 533.

[293] A. Somayazulu, S. Majumder, C. Chen, and K. Grauman, "ActiveRIR: Active audio-visual exploration for acoustic environment modeling," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 13 830–13 836.

[294] Y. Furletov, V. Willert, and J. Adamy, "Auditory scene understanding for autonomous driving," in *2021 IEEE Intelligent Vehicles Symposium (IV)*, 2021, pp. 697–702.

[295] V.-T. Tran and W.-H. Tsai, "Audio-vision emergency vehicle detection," *IEEE Sensors Journal*, vol. 21, no. 24, pp. 27 905–27 917, 2021.

[296] L. Banchero, F. Vacalebri-Lloret, J. M. Mossi, and J. J. Lopez, "Enhancing road safety with ai-powered system for effective detection and localization of emergency vehicles by sound," *Sensors*, vol. 25, no. 3, p. 793, 2025.

[297] A. L. Florentino, E. L. Diniz, and P. T. Aquino-Jr., "A dataset for environmental sound recognition in embedded systems for autonomous vehicles," *Scientific Data*, vol. 12, p. 1148, 2025.

[298] Z. Xing, H. Chen, B. Xie, J. Xu, Z. Guo, X. Xu, J. Hao, C.-W. Fu, X. Hu, and P.-A. Heng, "EchoTraffic: Enhancing traffic anomaly understanding with audio-visual insights," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 19 098–19 108.

[299] Z. Guo, F. Yang, X. Zhang, J. Guo, K. Zhao, Y. Zhou, P. Lu, S. Zheng, and Z. Zhang, "Listen, look, drive: Coupling audio instructions for user-aware vla-based autonomous driving," in *2026 IEEE Intelligent Vehicles Symposium (IV)*, 2026, pp. 981–988.

[300] NVIDIA, "Nemotron 3 Nano Omni: Efficient and open multimodal intelligence," 2026.

[301] ——, "Cosmos 3: Omnimodal world models for physical ai," *arXiv preprint arXiv:2606.02800*, 2026.

[302] H. Li, Y. Zhang, J. Zhu, S. Wang, M. A. Lee, H. Xu, E. Adelson, L. Fei-Fei, R. Gao, and J. Wu, "See, hear, and feel: Smart sensory fusion for robotic manipulation," in *Proceedings of The 6th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2023, pp. 1368–1378.

[303] A. Thankaraj and L. Pinto, "That sounds right: Auditory self-supervision for dynamic robot manipulation," in *Proceedings of The 7th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 1036–1049.

[304] J. Mejia, V. Dean, T. Hellebrekers, and A. Gupta, "Hearing touch: Audio-visual pretraining for contact-rich manipulation," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 6912–6919.

[305] R. Feng, D. Hu, W. Ma, and X. Li, "Play to the score: Stage-guided dynamic multi-sensory fusion for robotic manipulation," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 340–363.

[306] C. Higuera, A. Sharma, T. Fan, C. K. Bodduluri, B. Boots, M. Kaess, M. Lambeta, T. Wu, Z. Liu, F. R. Hogan, and M. Mukadam, "Tactile beyond pixels: Multisensory touch representations for robot manipulation," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 105–123.

[307] S. Li, J. Lu, Y. Song, X. Li, B. An, and P. Liu, "Hierarchical audio-visual-proprioceptive fusion for precise robotic manipulation," 2026.

[308] G. Ji, T. Li, D. Suh, Y. Zhang, B. Zhang, and N. Figueroa, "Dreaming the sound of contact: Leveraging video and audio generation for zero-shot force-aware manipulation," 2026, presented at the ICRA 2026 Workshop on Manipulation Robustness: Towards Human-Level Robustness under Real-World Challenges.

[309] C. Li, X. Chen, X. Ma, H. Sun, and B. Wang, "Skill acquisition and controller design of desktop robot manipulator based on audio–visual information fusion," *Machines*, vol. 10, no. 9, p. 772, 2022.

[310] D. Guo, H. Liu, and F. Sun, "Audio–visual language instruction understanding for robotic sorting," *Robotics and Autonomous Systems*, vol. 159, p. 104271, 2023.

[311] Z. Wang, G. Tian, and H. Pan, "Robot gaining robust pouring skills through fusing vision and audio," *ISA Transactions*, vol. 135, pp. 428–437, 2023.

[312] R. Wang, H. Geng, T. Li, P. Wu, F. Wang, G. Anumanchipalli, T. Darrell, B. Li, P. Abbeel, J. Malik, and A. A. Efros, "The sound of simulation: Learning multimodal sim-to-real robot policies with generative audio," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 420–436.

[313] K. Hiratsuka, B. Yen, and R. Kojima, "S2A2: Audio-visual imitation learning for manipulation tasks using acoustic spatial information," 2026.

[314] G. Skantze, A. Hjalmarsson, and C. Oertel, "Turn-taking, feedback and joint attention in situated human–robot interaction," *Speech Communication*, vol. 65, pp. 50–66, 2014.

[315] M. Shridhar and D. Hsu, "Interactive visual grounding of referring expressions for human-robot interaction," in *Proceedings of Robotics: Science and Systems*, Pittsburgh, Pennsylvania, 2018.

[316] Y. Chen, Q. Li, D. Kong, Y. L. Kei, S.-C. Zhu, T. Gao, Y. Zhu, and S. Huang, "YouRefIt: Embodied reference understanding with language and gesture," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 1385–1395.

[317] Y. Wang, K. Wang, Y. Wang, D. Guo, H. Liu, and F. Sun, "Audio-visual grounding referring expression for robotic manipulation," in *2022 IEEE International Conference on Robotics and Automation (ICRA)*, 2022, pp. 9258–9264.

[318] W. Wang, X. Li, Y. Dong, J. Xie, D. Guo, and H. Liu, "Natural language instruction understanding for robotic manipulation: A multisensory perception approach," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 9800–9806.

[319] G. A. Abbo and T. Belpaeme, "I was blind but now i see: Implementing vision-enabled dialogue in social robots," in *Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE, 2025, pp. 1176–1180.

[320] C. D. Hromei, F. Borazio, A. Sensi, E. Passone, D. Croce, and R. Basili, "Training multi-modal LLMs through dialogue planning for HRI," in *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics, 2025, pp. 16 266–16 284.

[321] S. Shrestha, Y. Zha, S. Banagiri, G. Gao, Y. Aloimonos, and C. Fermüller, "NatSGLD: A dataset with speech, gesture, logic, and demonstration for robot learning in natural human-robot interaction," in *Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction*, 2025, pp. 1093–1098.

[322] M. J. Pinto Bernal and T. Belpaeme, "Seeing only when needed: A multimodal architecture for context-aware social dialogue," in *Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*. ACM, 2026, pp. 701–705.

[323] A. Ekrekli, A. Angleraud, G. Sharma, and R. Pieters, "Co-speech gestures for human-robot collaboration," in *2023 Seventh IEEE International Conference on Robotic Computing (IRC)*. IEEE, 2023, pp. 110–114.

[324] G. Campagna, C. Frommel, T. Haase, A. Gottardi, E. Villagrossi, D. Chrysostomou, and M. Rehm, "Fostering trust through gesture and voice-controlled robot trajectories in industrial human-robot collaboration," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 11 853–11 859.

[325] R. Ojanen and R. Pieters, "Bidirectional verbal communication in human-robot collaboration," in *IFAC Conference on Manufacturing Modelling, Management and Control*. IFAC, 2025, pp. 2274–2279.

[326] J. Wang, C. Wang, R. S. Zarrin, and Z. Erickson, "Bidirectional human-robot communication for physical human-robot interaction," in *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*. Association for Computing Machinery, 2026, pp. 572–580.

[327] O. Tasneem and R. Pieters, "Human–robot collaborative visual inspection with large language models," *Robotics and Computer-Integrated Manufacturing*, vol. 98, p. 103154, 2026.

[328] A. Yu, C. Li, L. Macesanu, A. Balaji, R. Ray, R. Mooney, and R. Martín-Martín, "Mixed-initiative dialog for human-robot collaborative manipulation," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[329] B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, Q. Vuong, V. Vanhoucke, H. Tran, R. Soricut, A. Singh, J. Singh, P. Sermanet, P. R. Sanketi, G. Salazar, M. S. Ryoo, K. Reymann, K. Rao, K. Pertsch, I. Mordatch, H. Michalewski, Y. Lu, S. Levine, L. Lee, T.-W. E. Lee, I. Leal, Y. Kuang, D. Kalashnikov, R. Julian, N. J. Joshi, A. Irpan, B. Ichter, J. Hsu, A. Herzog, K. Hausman, K. Gopalakrishnan, C. Fu, P. Florence, C. Finn, K. A. Dubey, D. Driess, T. Ding, K. M. Choromanski, X. Chen, Y. Chebotar, J. Carbajal, N. Brown, A. Brohan, M. G. Arenas, and K. Han, "RT-2: Vision-language-action models transfer web knowledge to robotic control," in *Proceedings of the 7th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 2165–2183.

[330] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. P. Foster, P. R. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, "OpenVLA: An open-source vision-language-action model," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 2679–2713.

[331] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, L. X. Shi, L. Smith, J. Tanner, Q. Vuong, A. Walling, H. Wang, and U. Zhilinsky, "π_0: A vision-language-action flow model for general robot control," in *Proceedings of Robotics: Science and Systems*, 2025.

[332] J. Jones, O. Mees, C. Sferrazza, K. Stachowicz, P. Abbeel, and S. Levine, "Beyond sight: Finetuning generalist robot policies with heterogeneous sensors via language grounding," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 5961–5968.

[333] X. Wei, H. Zhang, X. Cao, S. Xie, W. Ge, Y. Li, and C. Wang, "Audio-VLA: Adding contact audio perception to vision-language-action model for robotic manipulation," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026, arXiv:2511.09958.

[334] L. Li, J. Fan, X. Ni, S. Qin, W. Li, and F. Gao, "SVA: Towards speech-enabled vision-language-action model," *Pattern Recognition*, vol. 173, p. 112915, 2026.

[335] C. Nie, T. Deng, G. Wang, Z. Liu, and H. Wang, "Towards the vision-sound-language-action paradigm: The HEAR framework for sound-centric manipulation," *The International Journal of Robotics Research*, 2026, accepted for publication.

[336] Open X-Embodiment Collaboration, "Open x-embodiment: Robotic learning datasets and RT-X models," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 6892–6903.

[337] D. Ghosh, H. R. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu, J. Luo, Y. L. Tan, L. Y. Chen, Q. Vuong, T. Xiao, P. R. Sanketi, D. Sadigh, C. Finn, and S. Levine, "Octo: An open-source generalist robot policy," in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, 2024.

[338] Y. Hu, Y. Guo, P. Wang, X. Chen, Y.-J. Wang, J. Zhang, K. Sreenath, C. Lu, and J. Chen, "Video prediction policy: A generalist robot policy with predictive visual representations," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 24 328–24 346.

[339] P. Atreya, K. Pertsch, T. Lee, M. J. Kim, A. Jain, A. Kuramshin, C. Neary, E. S. Hu, K. Arora, K. Ellis, L. Macesanu, M. Leonard, M. Cho, O. Aslan, S. Dass, T. Wang, X. Yuan, A. Gupta, D. Jayaraman, G. Berseth, K. Daniilidis, R. Martín-Martín, Y. Lee, P. Liang, C. Finn, and S. Levine, "RoboArena: Distributed real-world evaluation of generalist robot policies," in *Proceedings of the 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 336–364.

[340] Y. Xing, X. Luo, J. Xie, L. Gao, H. T. Shen, and J. Song, "Shortcut learning in generalist robot policies: The role of dataset diversity and fragmentation," in *Proceedings of the 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 3239–3266.

[341] S.-B. Kim, H.-B. Oh, J.-M. Lee, A. Senocak, J. S. Chung, and T.-H. Oh, "AVHBench: A cross-modal hallucination benchmark for audio-visual large language models," in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: https://openreview.net/forum?id=jTEKTdI3K9

[342] D. Hu, R. Qian, M. Jiang, X. Tan, S. Wen, E. Ding, W. Lin, and D. Dou, "Discriminative sounding objects localization via self-supervised audiovisual matching," *Advances in Neural Information Processing Systems*, vol. 33, pp. 10 077–10 087, 2020.

[343] S. Mo and P. Morgado, "A closer look at weakly-supervised audio-visual source localization," *Advances in Neural Information Processing Systems*, vol. 35, pp. 37 524–37 536, 2022.

[344] X. Hu, Z. Chen, and A. Owens, "Mix and localize: Localizing sound sources in mixtures," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2022, pp. 10 473–10 482.

[345] M. Hamilton, A. Zisserman, J. R. Hershey, and W. T. Freeman, "Separating the "chirp" from the "chat": Self-supervised visual grounding of sound and language," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 13 117–13 127.

[346] A. Senocak, H. Ryu, J. Kim, T.-H. Oh, H. Pfister, and J. S. Chung, "Aligning sight and sound: Advanced sound source localization through audio-visual alignment," *arXiv preprint arXiv:2407.13676*, 2024.

[347] J. Zhou, X. Shen, J. Wang, J. Zhang, W. Sun, J. Zhang, S. Birchfield, D. Guo, L. Kong, M. Wang *et al.*, "Audio-visual segmentation with semantics," *International Journal of Computer Vision*, vol. 133, no. 4, pp. 1644–1664, 2025.

[348] J. Liu, Y. Wang, C. Ju, C. Ma, Y. Zhang, and W. Xie, "Annotation-free audio-visual segmentation," in *2024 IEEE/CVF Winter Conference on*

TABLE V
REPRESENTATIVE BENCHMARKS AND DERIVED EVALUATION PROTOCOLS FOR PIXEL GROUNDING. THE BENCHMARKS COVER SINGLE- AND MULTI-SOURCE LOCALIZATION, SEMANTIC AND INSTANCE SEGMENTATION, REFERRING GROUNDING, ROBUSTNESS, AND CONTINUAL SETTINGS.

Benchmark	Year	Task / Main Challenge	Scale / Categories	Annotation	Main Evaluation
Flickr-SoundNet / SSL [95]	2018	Single-source response-map localization	5,000 annotated image–sound pairs; 2,786 consensus pairs; 250-sample test set	Bounding boxes from three annotators	cIoU (LocAcc), AUC
MUSIC-Synthetic [342]	2020	Sounding-versus-silent discrimination in mixtures	Four visual panels; two sounding instruments	Concatenated frames and mixed audio	IoU, AUC
VGG-Sound Source (VGG-SS) [20]	2021	Diverse sound-source localization	5,158 clips, 220 categories	Verified sounding-object boxes on center frames	cIoU (LocAcc), AUC
Extended Flickr-SoundNet / Extended VGG-SS [343]	2022	Absent-source and mismatch robustness	500 / 10,316 positive–negative pairs	Inherited boxes with real and constructed negatives	AP, max-F1, LocAcc
VGGSound-Instruments / VoxCeleb2 protocols [344]	2022	Multi-instrument and multi-speaker localization	VGGSound-Instruments: 32K clips, 37 classes; 446 mask-annotated test frames; VoxCeleb2: 1K mask-annotated test samples	Object / face segmentation masks; synthetic multi-source mixtures	AP, IoU, AUC
VGGSound-Single / VGGSound-Duet [99]	2023	General-category single- and multi-source localization	150K training clips, 221 classes; 5,158 Single test samples; 5,158 synthetic Duet samples	VGG-SS boxes; Duet formed by concatenating two test frames and mixing their audio	AP, IoU, AUC
ADE20K-Speech / ADE20K-Sound [345]	2024	Speech- and sound-conditioned grounding	3,030 speech pairs; 106 sound pairs	Semantic masks paired with speech or sounds	mAP, mIoU
IS3 [346]	2024	Interactive sound-source localization and cross-modal interaction	3,240 two-object images, 118 categories; 6,480 object instances	Bounding boxes and segmentation masks	cIoU, AUC; IIoU, IAUC and adaptive variants
Extended-IS3 [103]	2025	Simultaneous grounding of speech and non-speech sound	3,420 images; two visible objects with sound and generated speech cues	Object masks with object-specific sound and speech samples	mAP, mIoU
AVSBench [21]	2022	Binary AVS; S4 single-source / MS3 multi-source	5,356 videos, 23 classes	Binary sounding-object masks	mIoU, $\mathcal{F}$
AVSBench-Semantic [347]	2023	Audio-visual semantic segmentation	12,356 videos, 70 classes	Semantic pixel masks	mIoU, $\mathcal{F}$
AVS-Synthetic [348]	2024	Synthetic AVS training data	58,405 images, 62,609 masks, 46 classes	Image–audio–mask triplets	mIoU, F-score
LU-AVS [108]	2024	Long-untrimmed spatial-temporal grounding	7,257 videos, 88 categories; 10.35M masks and 10.98M bounding boxes	Frame-level sounding intervals; masks for 6,627 normal videos and bounding boxes for all 7,257 videos	m_{tIoU}, m_{tF}, m_{stIoU}, m_{sF}
VPO-SS/MS/MSMI [109]	2024	Visual bias; multi-source / multi-instance grounding	21 COCO classes	Semantic masks with synthetic AV pairs	mIoU, F-score, FDR
AVSBench-OV [349]	2024	Open-vocabulary AV semantic segmentation	40 base + 30 novel classes	Semantic pixel masks	Base / novel / overall mIoU; harmonic mIoU
Ref-AVSBench [110]	2024	Referring AVS with multimodal cues	4,002 videos, 20,261 expressions	Target masks with referring expressions	$\mathcal{J}$, $\mathcal{F}$, null-reference evaluation
AVISeg [111]	2025	Instance-level AVS and temporal tracking	926 videos, 26 categories, 94,074 masks	Instance masks, identities, and sounding states	mAP, HOTA, FSLA
OmniAVS [113]	2025	Omnimodal referring and reasoning	2,104 videos, 61,095 expressions	Masks, expressions, and explanations	$\mathcal{J}\&\mathcal{F}$ (masks), METEOR (text)
AVSBench-CI/CIS/CIM [350]	2025	Class-incremental AVS	70 categories with incremental splits	Semantic masks with incremental splits	mIoU and incremental metrics
AVSBench-Robust [351]	2026	Robustness to invalid audio evidence	Based on AVSBench S4 / MS3 / AVSS	Positive and negative AV pairs	mIoU, $\mathcal{F}$, FPR, G-mIoU, G-$\mathcal{F}$, G-FPR
R^2-AVSBench [352]	2026	Reasoning-intensive referring AVS	400 videos	Target masks with indirect references	$\mathcal{J}$, $\mathcal{F}$
CL-AVS [353]	2026	Exemplar-free continual AVS	SS-AVS / MS-AVS streams	Existing AVS annotations	mAP, Max-F; LA, AA, Forgetting, BWT, FWT

Applications of Computer Vision (WACV). IEEE, 2024, pp. 5592–5602.

[349] R. Guo, L. Qu, D. Niu, Y. Qi, W. Yue, J. Shi, B. Xing, and X. Ying, "Open-vocabulary audio-visual semantic segmentation," in *Proceedings of the 32nd ACM International conference on multimedia*, 2024, pp. 7533–7541.

[350] Y. Hong, Q. Yang, T. Zhang, Z. Wang, Z. Fu, K. Ding, B. Fan, and S. Xiang, "Taming modality entanglement in continual audio-visual segmentation," *arXiv preprint arXiv:2510.17234*, 2025.

[351] J. Li, W. Zhao, Z. Huang, Y. Guo, and Y. Tian, "Do audio-visual segmentation models truly segment sounding objects?" in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 8, 2026, pp. 6172–6180.

[352] J. Zhou, Y. Zhou, M. Han, T. Wang, X. Chang, H. Cholakkal, and R. M. Anwer, "Think before you segment: An object-aware reasoning agent for referring audio-visual segmentation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 16, 2026, pp. 13 665–13 673.

[353] S. Raghavan, G. Vinod, B. Coburn, and F. Zhu, "Can you hear, localize, and segment continually? an exemplar-free continual learning benchmark for audio-visual segmentation," *arXiv preprint arXiv:2603.08967*, 2026.

APPENDIX

A. More Benchmark

Pixel Grounding Benchmark. Pixel-grounding benchmarks reflect the same progression in spatial granularity and scene complexity. Flickr-SoundNet and VGG-SS mainly evaluate single-source localization, while later protocols introduce mixtures, silent or absent sources, multi-speaker and multi-instrument scenes, and cross-modal mismatches. Dense AVS benchmarks further extend evaluation from binary segmentation to semantic, long-video, open-vocabulary, referring, instance-level, omnimodal, continual-learning, and robustness settings. Representative datasets, annotation types, and evaluation metrics are summarized in Table V.